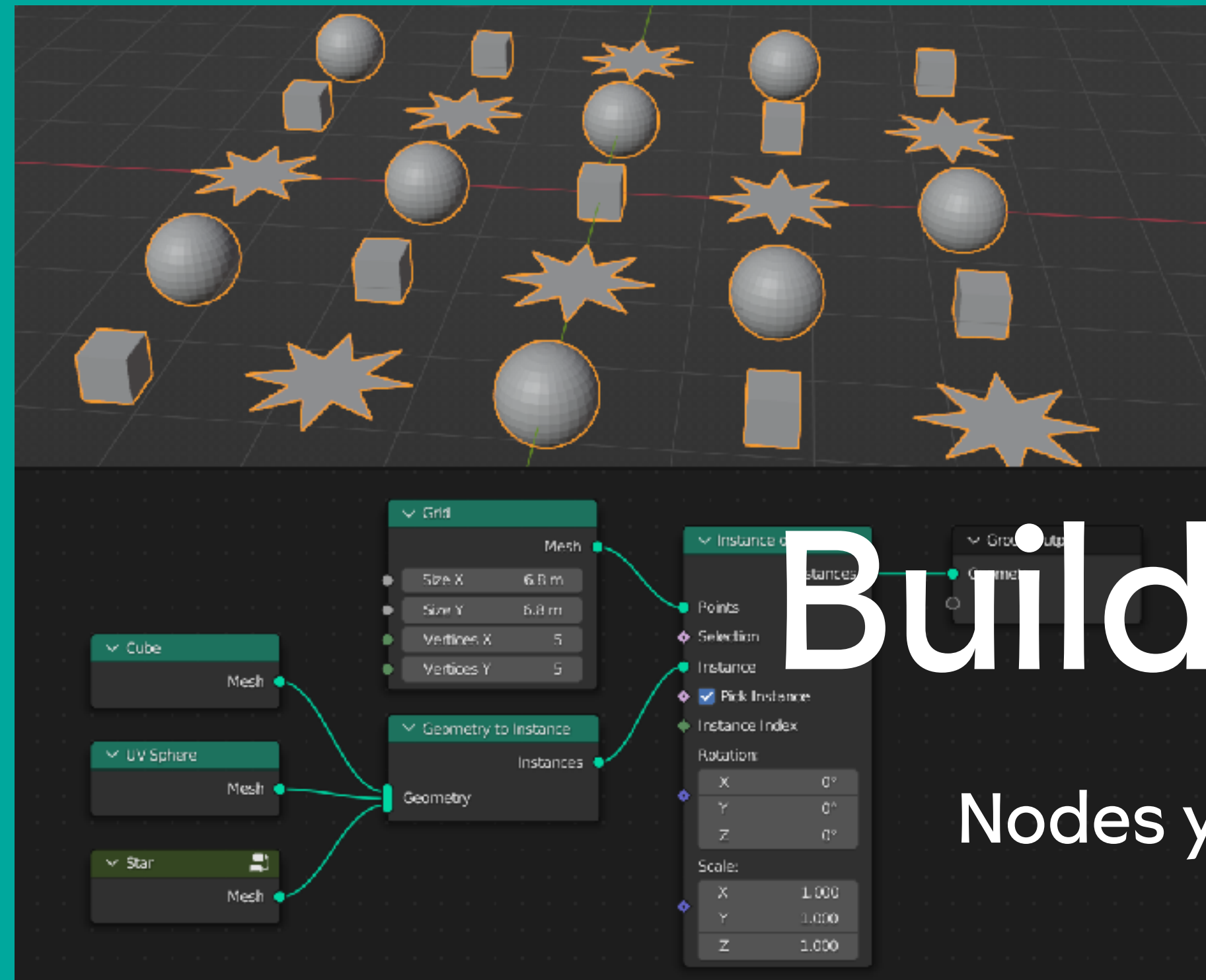




Building Blocks

Nodes you have learned so far

All Nodes (you have learned so far)

Attribute Statistic Node
Domain Size Node
Blur Attribute Node
Capture Attribute Node
Remove Named Attribute Node
Store Named Attribute Node
Boolean Node
Color Node
Image Node
Integer Node
Material Node
String Node
Value Node
Vector Node
Collection Info Node
Image Info Node
Is Viewport Node
Object Info Node
Self Object Node
Viewer Node
ID Node
Index Node
Named Attribute Node
Normal Node
Position Node
Radius Node
Geometry Proximity Node
Index of Nearest
Raycast Node
Sample Index Node
Sample Nearest Node
Set ID Node
Set Position Node

Bounding Box Node
Convex Hull Node
Delete Geometry Node
Duplicate Elements Node
Merge by Distance Node
Transform Geometry Node
Separate Components Node
Separate Geometry Node
Join Geometry Node
Geometry to Instance Node
Curve Handle Position Node
Curve Length Node
Curve Tangent Node
Curve Tilt Node
Endpoint Selection Node
Handle Type Selection Node
Is Spline Cyclic Node
Spline Length Node
Spline Parameter Node
Spline Resolution Node
Sample Curve Node
Set Curve Normal Node
Set Curve Radius Node
Set Curve Tilt Node
Set Handle Positions Node
Set Handle Type Node
Set Spline Cyclic Node
Set Spline Resolution Node
Set Spline Type Node
Curve to Mesh Node
Curve to Points Node
Deform Curves on Surface Node
Fill Curve Node

Fillet Curve Node
Interpolate Curves Node
Resample Curve Node
Reverse Curve Node
Subdivide Curve Node
Trim Curve Node
Arc Node
Bézier Segment Node
Curve Circle Node
Curve Line Node
Curve Spiral Node
Quadratic Bézier Node
Quadrilateral Node
Star Node
Curve of Point Node
Offset Point in Curve Node
Points of Curve Node
Instance on Points Node
Instances to Points Node
Rotate Instances Node
Scale Instances Node
Translate Instances Node
Realize Instances Node
Instance Rotation Node
Instance Scale Node
Edge Angle Node
Edge Neighbors Node
Edge Vertices Node
Edges to Face Groups Node
Face Area Node
Face Neighbors Node
Face Group Boundaries Node
Is Face Planar Node

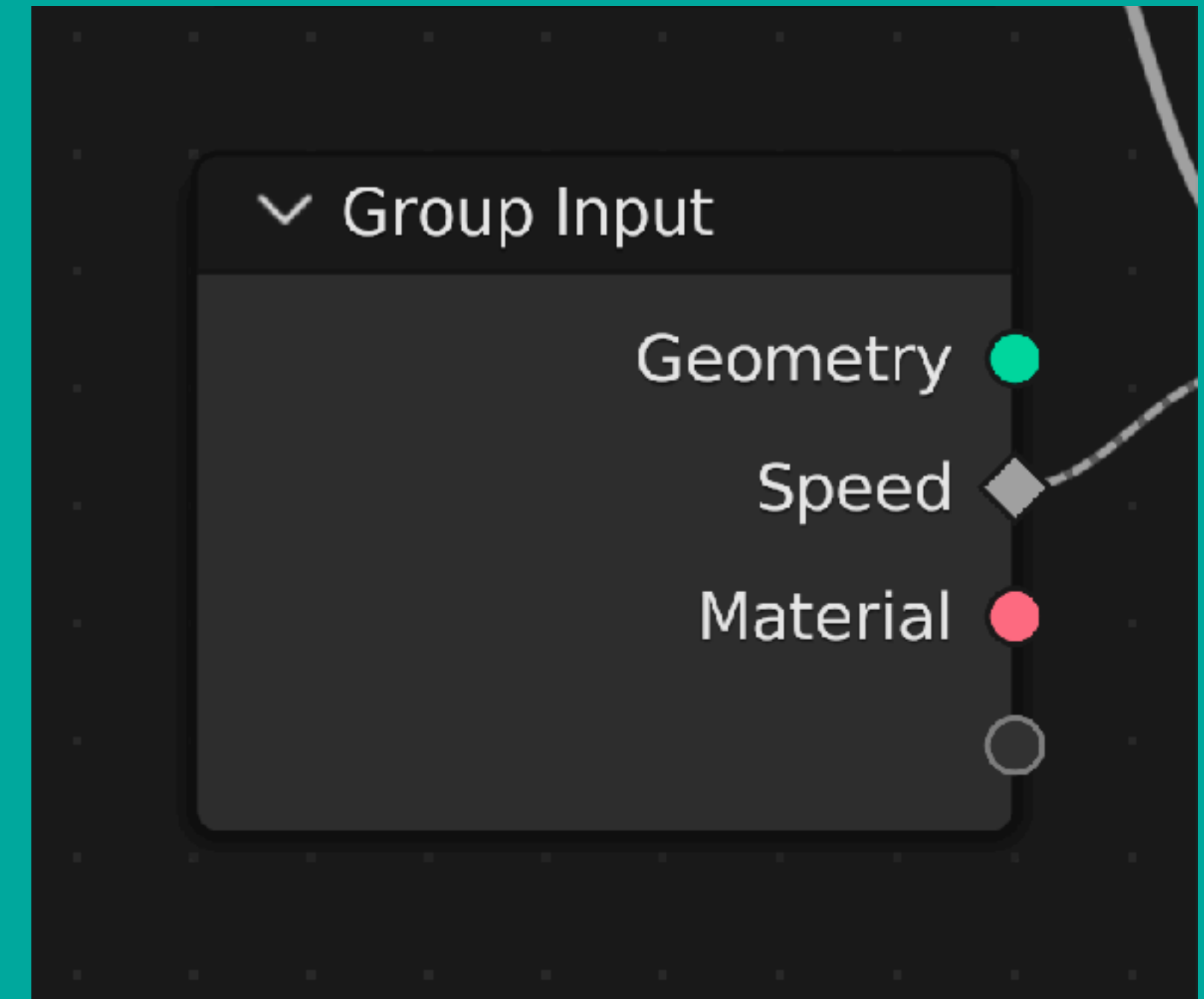
Is Shade Smooth Node
Mesh Island Node
Shortest Edge Paths Node
Vertex Neighbors Node
Sample Nearest Surface Node
Sample UV Surface Node
Set Shade Smooth Node
Dual Mesh Node
Edge Paths to Curves Node
Edge Paths to Selection Node
Extrude Mesh Node
Flip Faces Node
Mesh Boolean Node
Mesh to Curve Node
Mesh to Points Node
Mesh to Volume Node
Scale Elements Node
Split Edges Node
Subdivide Mesh Node
Subdivision Surface Node
Triangulate Node
Cone Node
Cube Node
Cylinder Node
Grid Node
Icosphere Node
Mesh Circle Node
Mesh Line Node
UV Sphere Node
Corners of Face Node
Corners of Vertex Node
Edges of Corner Node
Edges of Vertex Node

Face of Corner Node
Offset Corner in Face Node
Vertex of Corner Node
Pack UV Islands Node
UV Unwrap Node
Distribute Points in Volume
Distribute Points on Faces
Points Node
Points to Vertices Node
Points to Volume Node
Set Point Radius Node
Volume Cube Node
Volume to Mesh Node
Simulation Zone
Replace Material Node
Material Index Node
Material Selection Node
Set Material Node
Set Material Index Node
Brick Texture Node
Checker Texture Node
Gradient Texture Node
Image Texture Node
Magic Texture Node
Musgrave Texture Node
Noise Texture Node
Voronoi Texture Node
Wave Texture Node
White Noise Texture Node
Color Ramp Node
Combine Color Node
Mix Node
RGB Curves Node

Separate Color Node
Join Strings Node
Replace String Node
Slice String Node
Special Characters Node
String Length Node
String to Curves Node
Value to String Node
Combine XYZ Node
Separate XYZ Node
Vector Curves Node
Vector Math Node
Vector Rotate Node
Accumulate Field Node
Evaluate at Index Node
Evaluate on Domain Node
Boolean Math Node
Clamp Node
Compare Node
Float Curve
Float To Integer Node
Map Range Node
Math Node
Mix Node
Align Euler to Vector Node
Rotate Euler Node
Random Value Node
Switch Node
Group

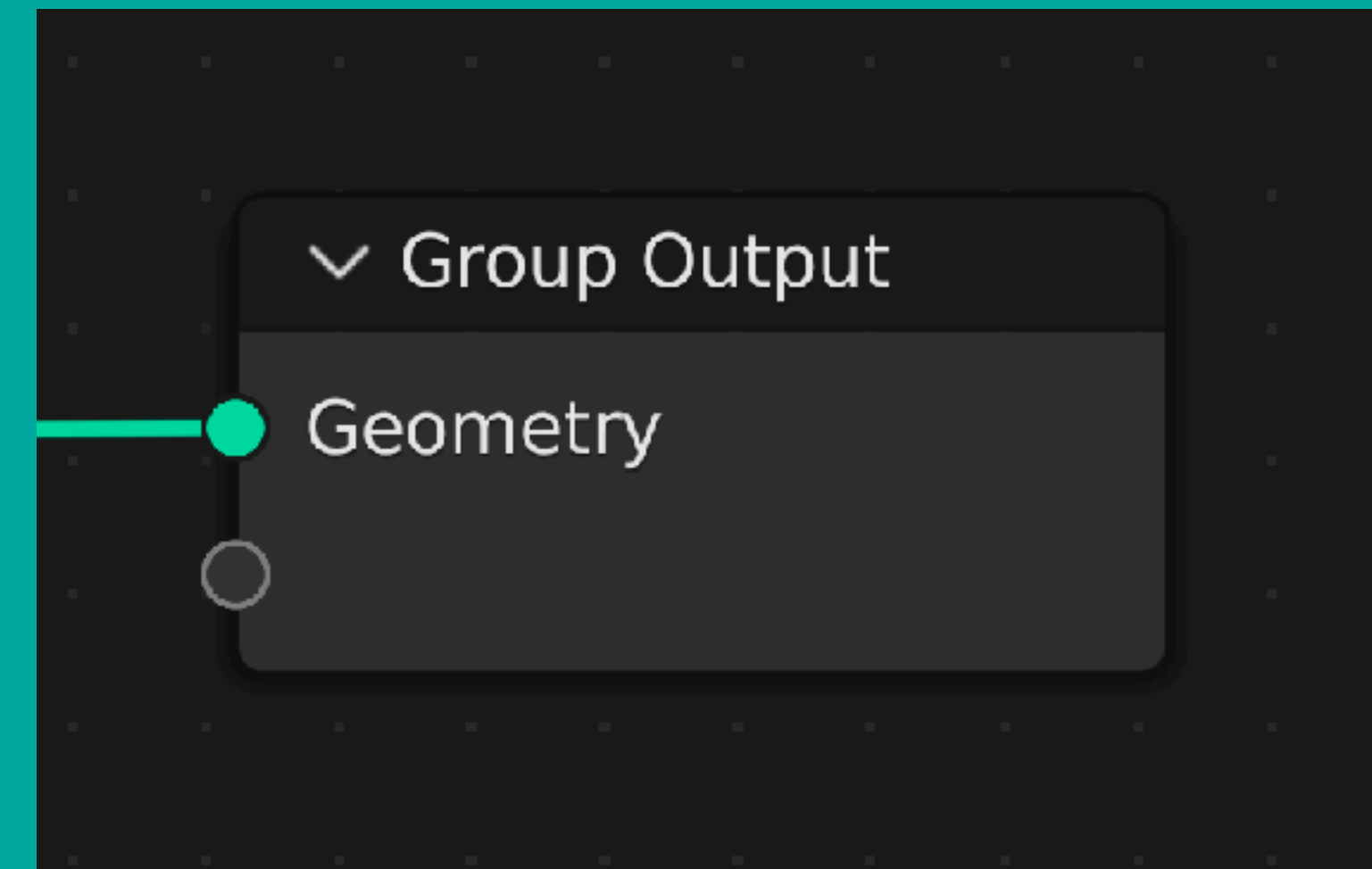
Group Input

- Add object from viewport to the node tree (Geometry)
- Add information from outside to the node tree (your own group inputs, like Seed, Speed or Material)



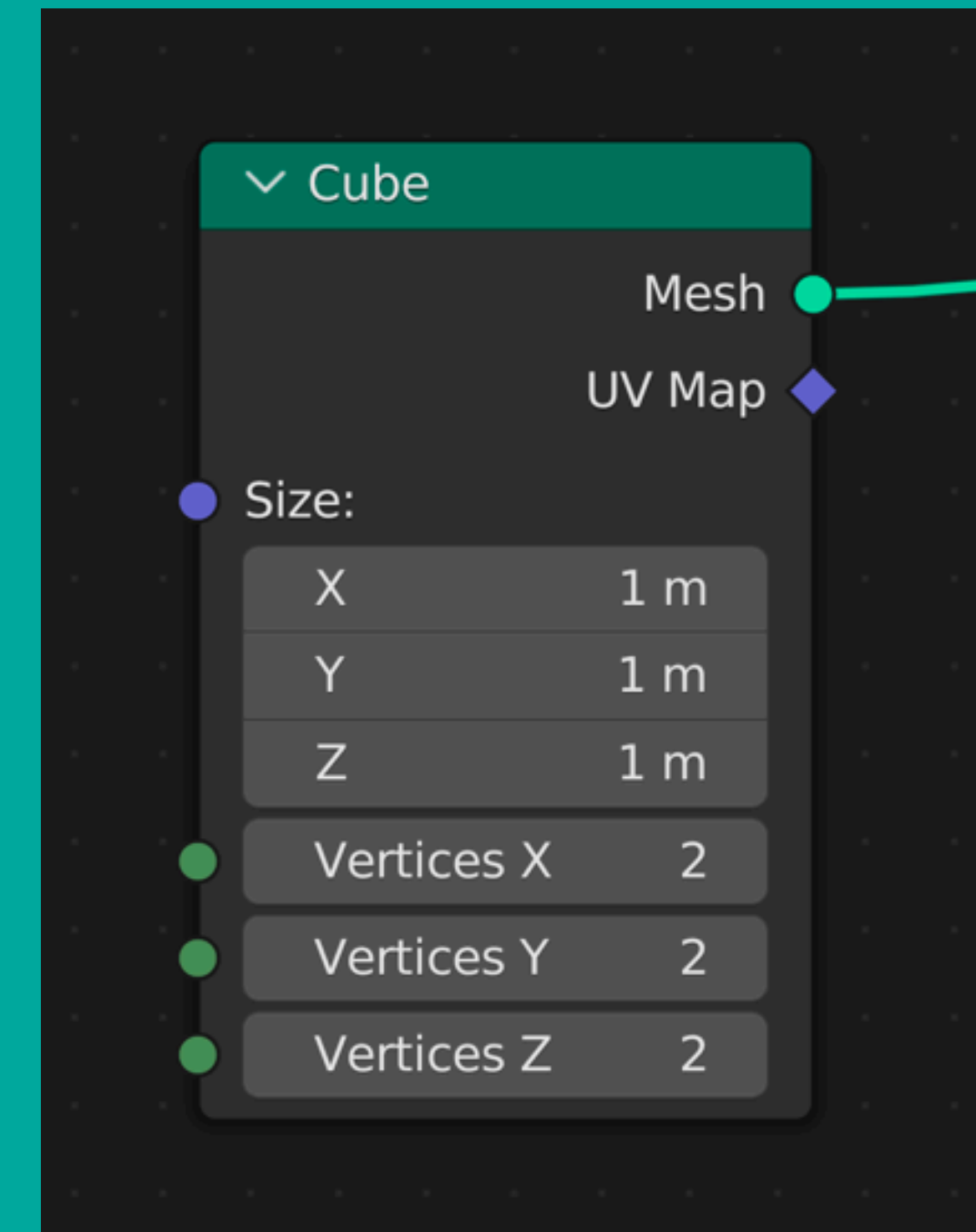
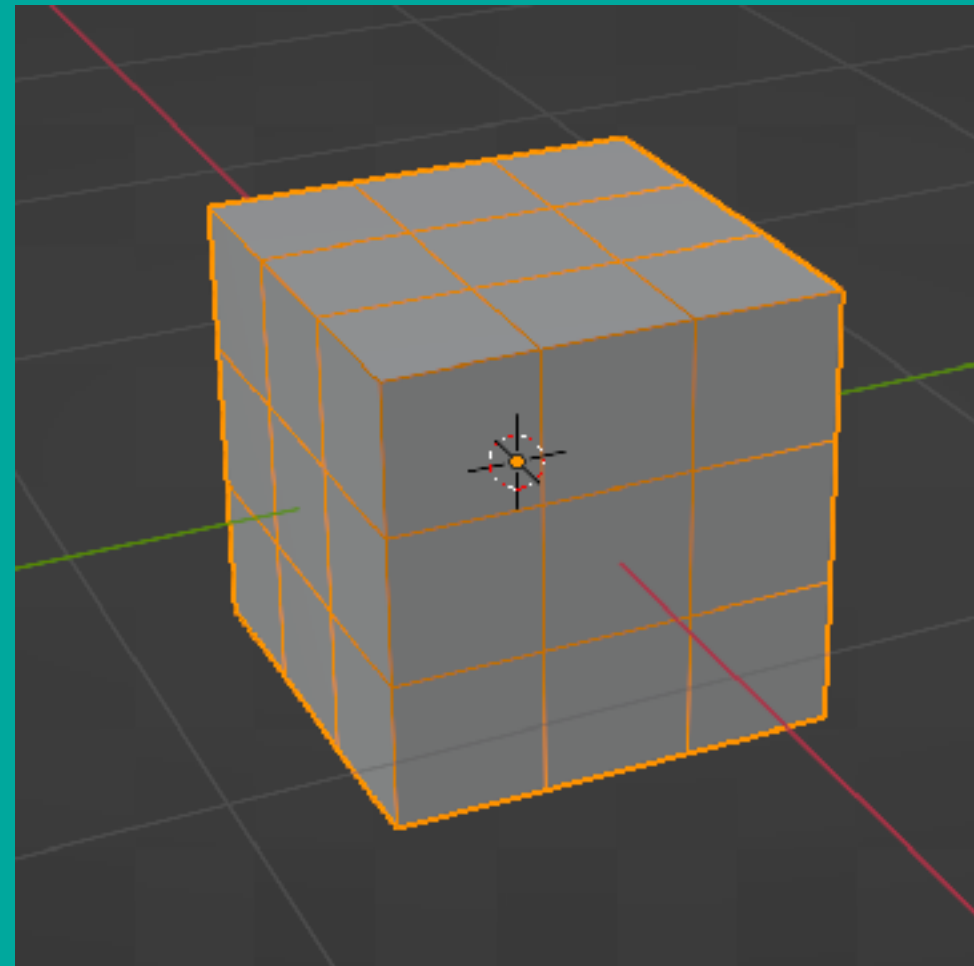
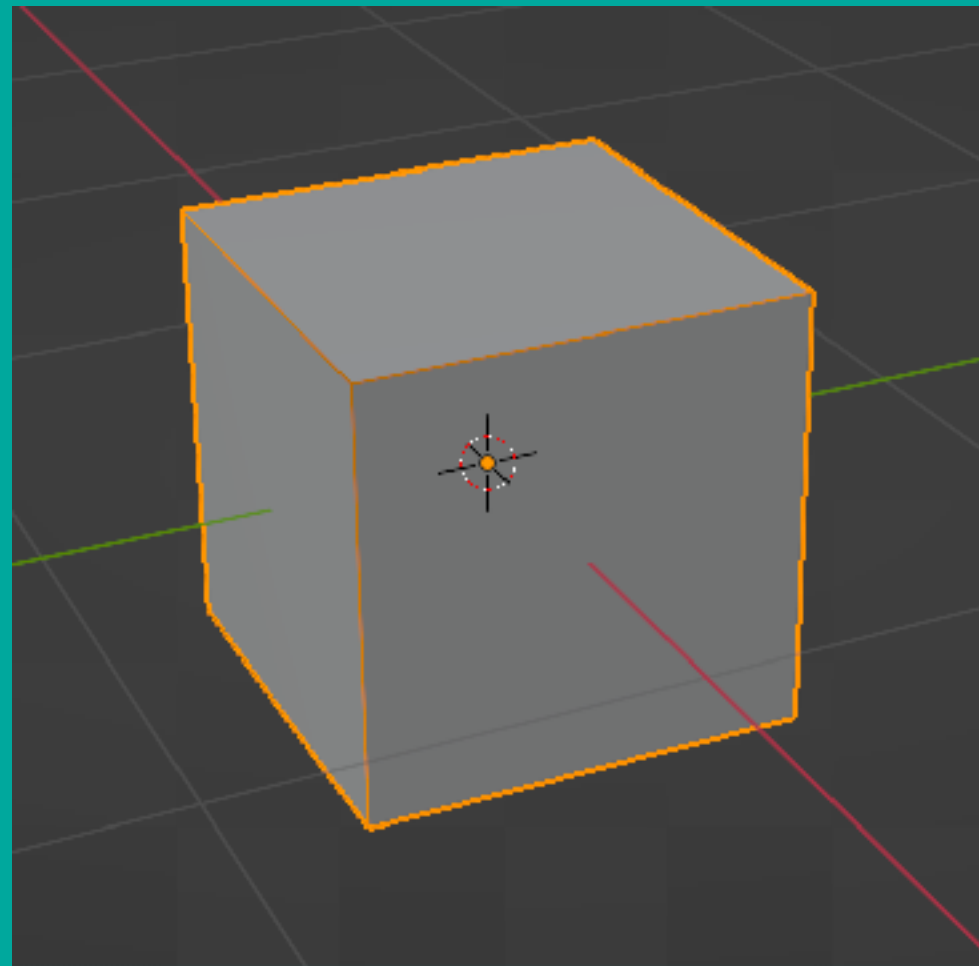
Group Output

- Outputs information from node tree
- You always use this, otherwise your Geometry Nodes don't generate any output
- You can only have 1 group output node!



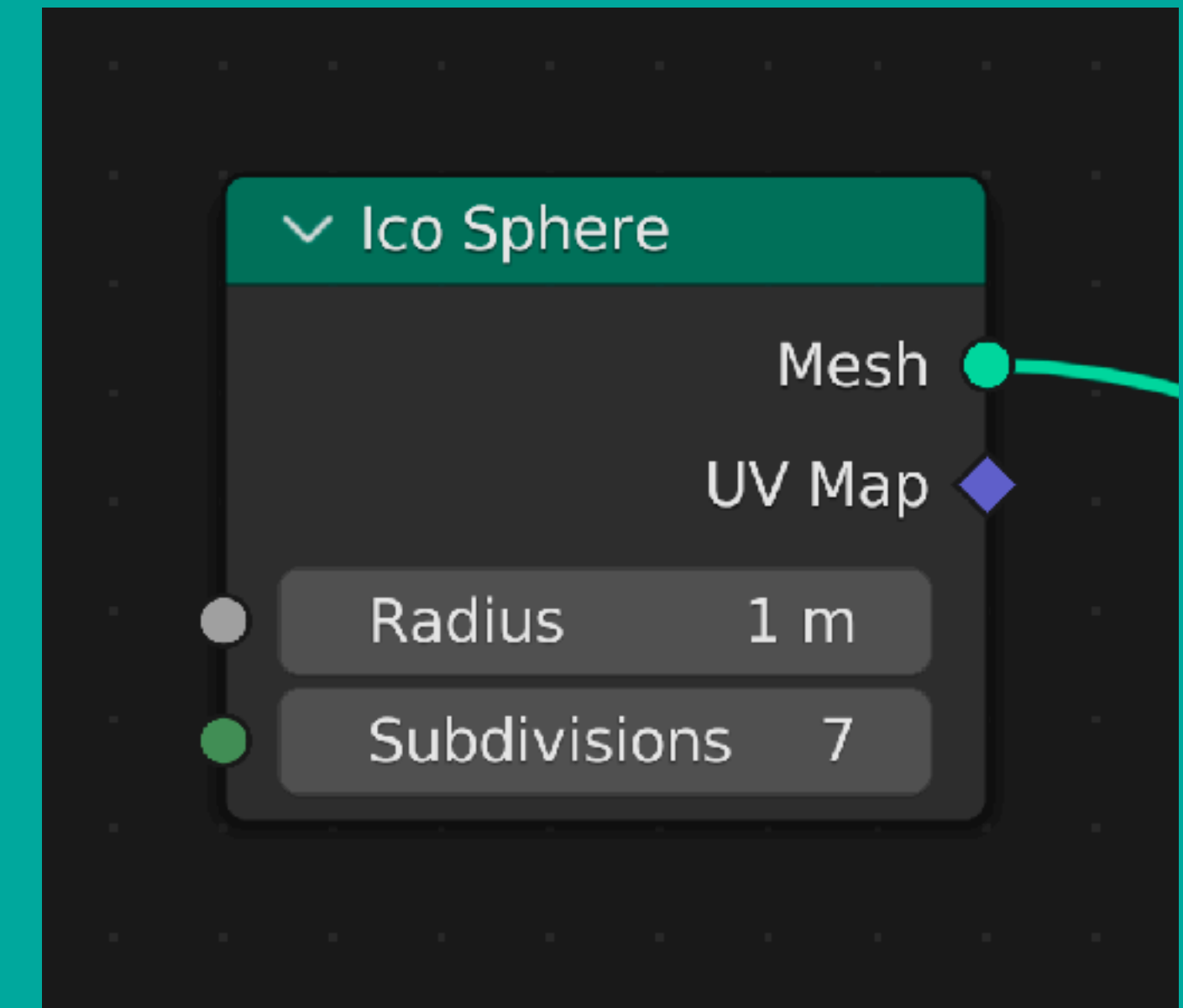
Cube

- Makes a cube
- Size sets size
- Vertices set subdivisions; the example on the left has 2x2x2 vertices, on the right has 4x4x4



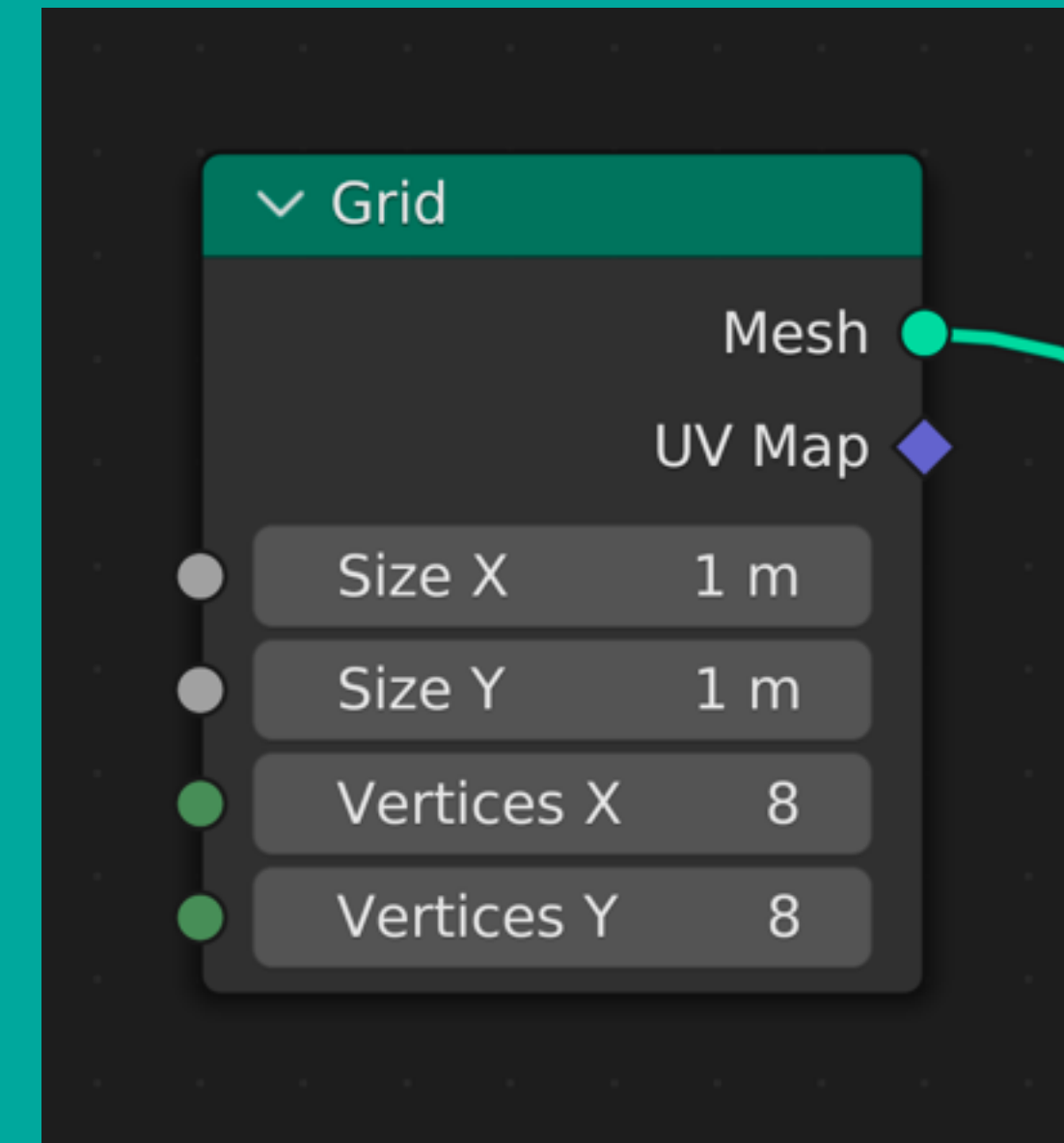
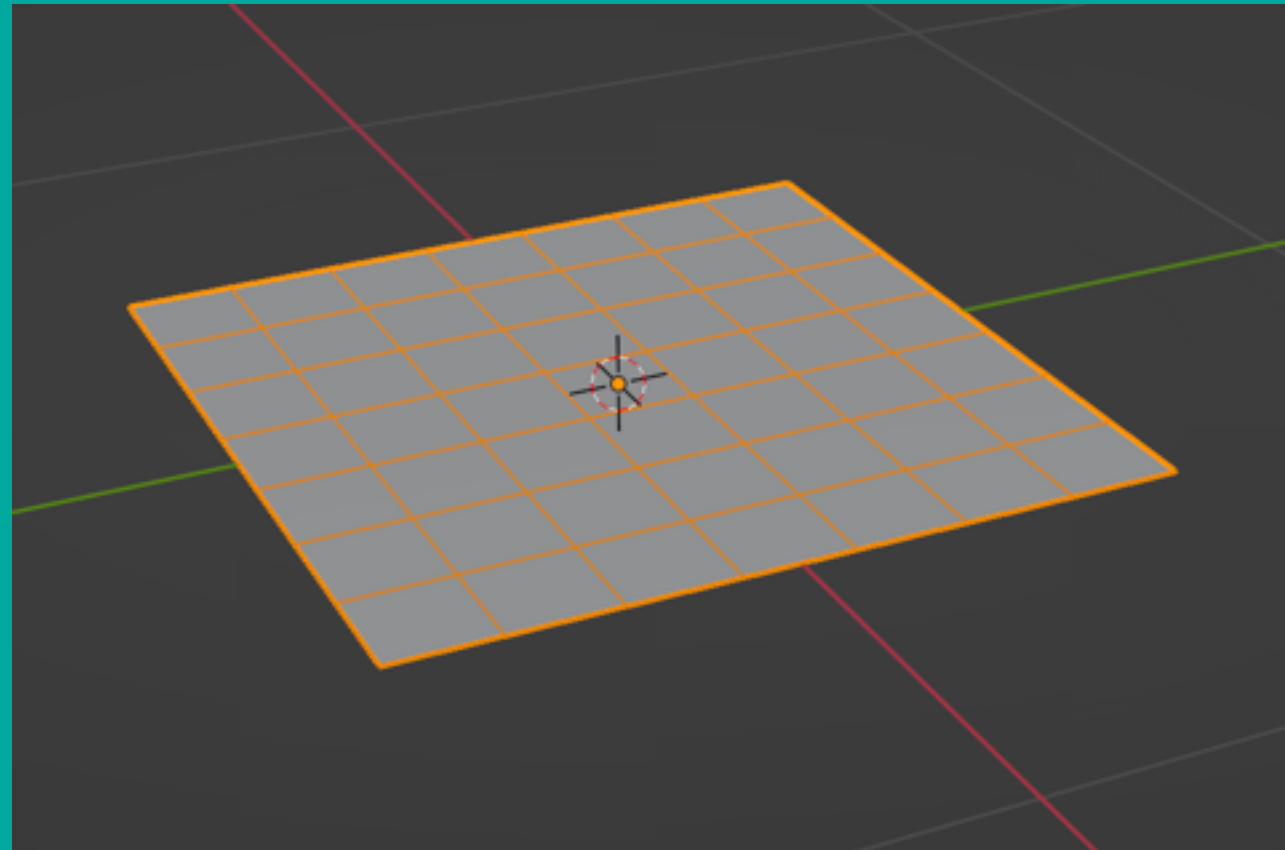
Ico Sphere

- Creates an Ico Sphere
- Radius sets the size
- Subdivisions sets the smoothness
- Usually don't set this higher than 7, or your geometry might crash Blender!
- Is different from the UV sphere:



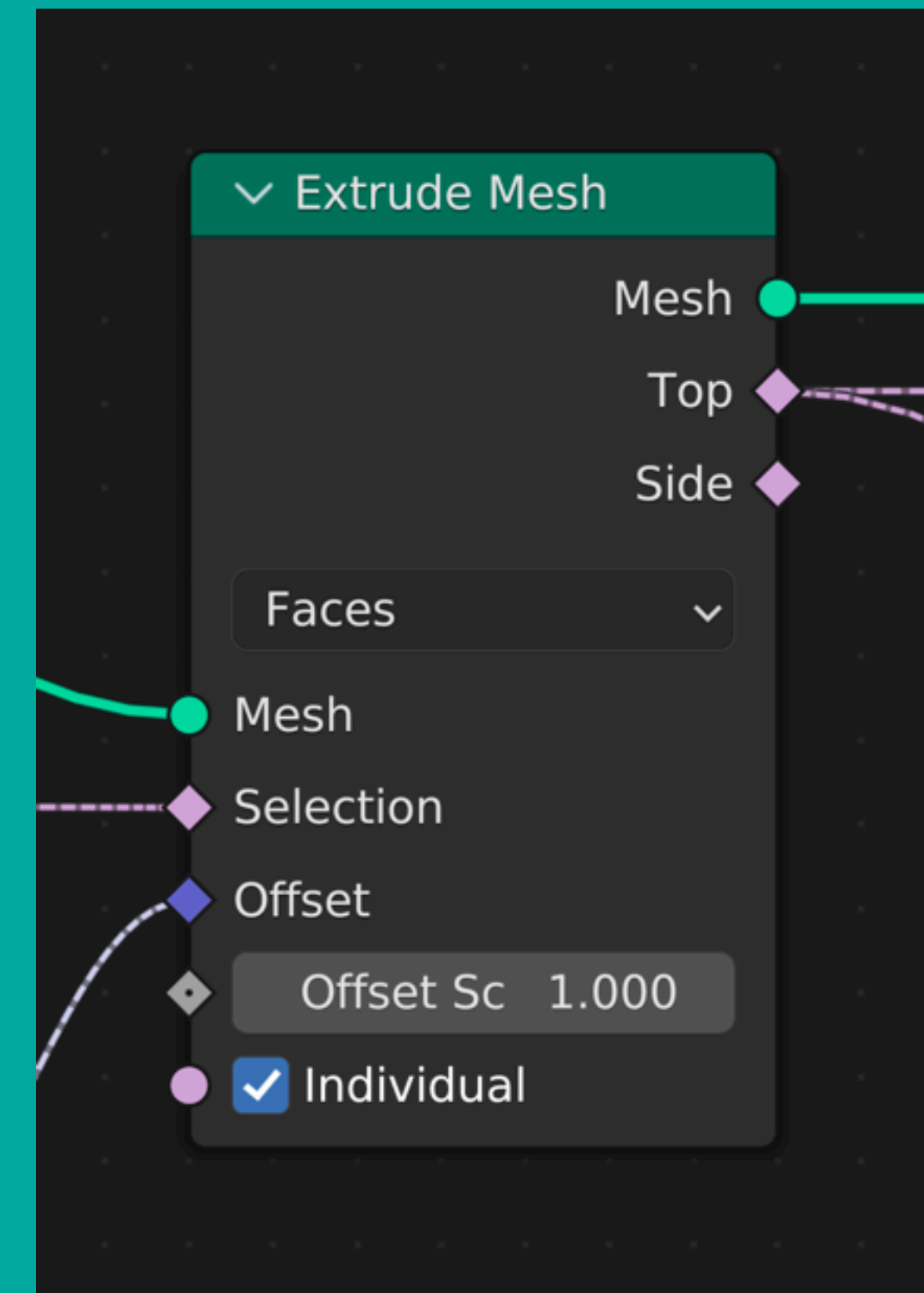
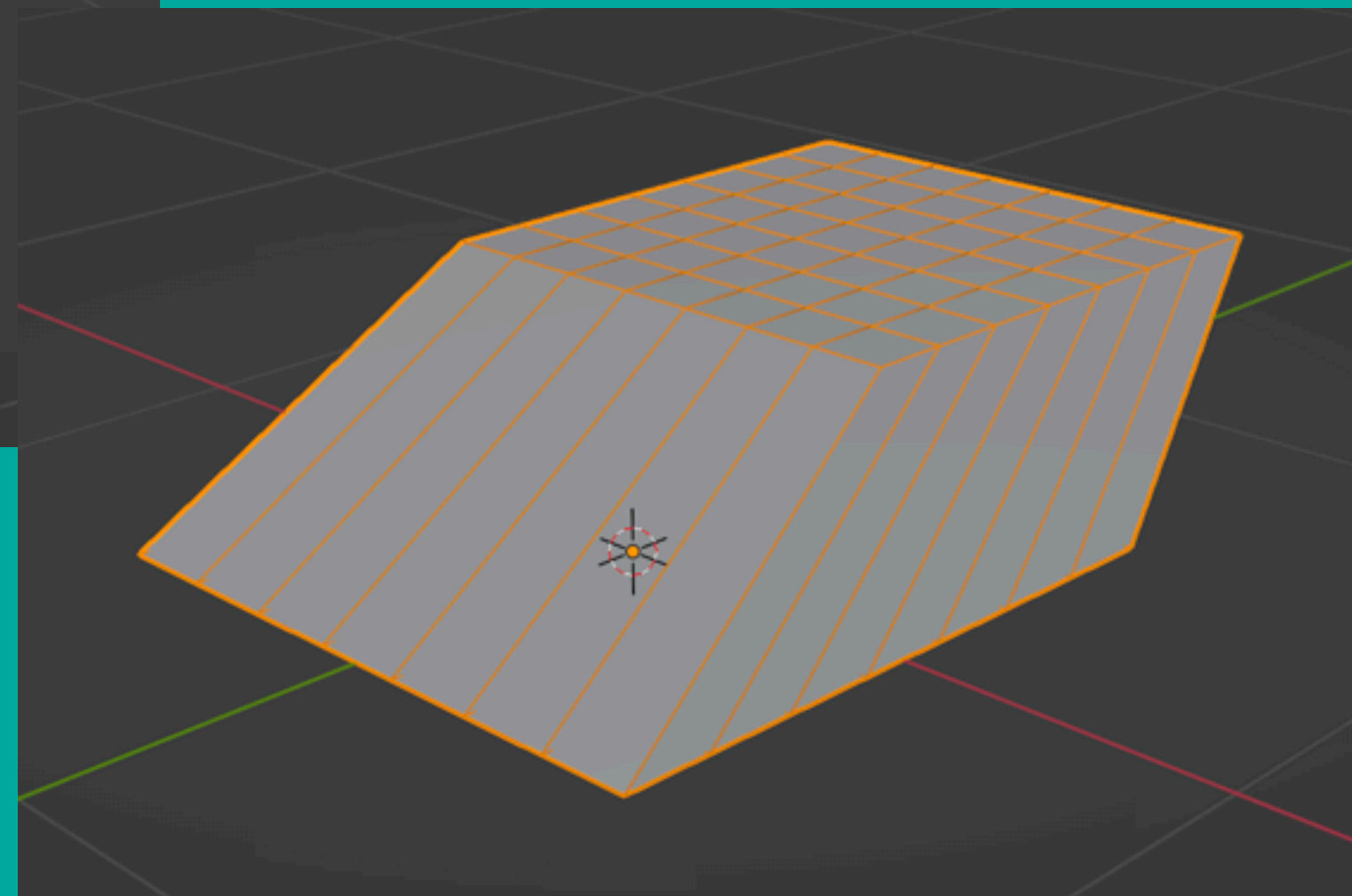
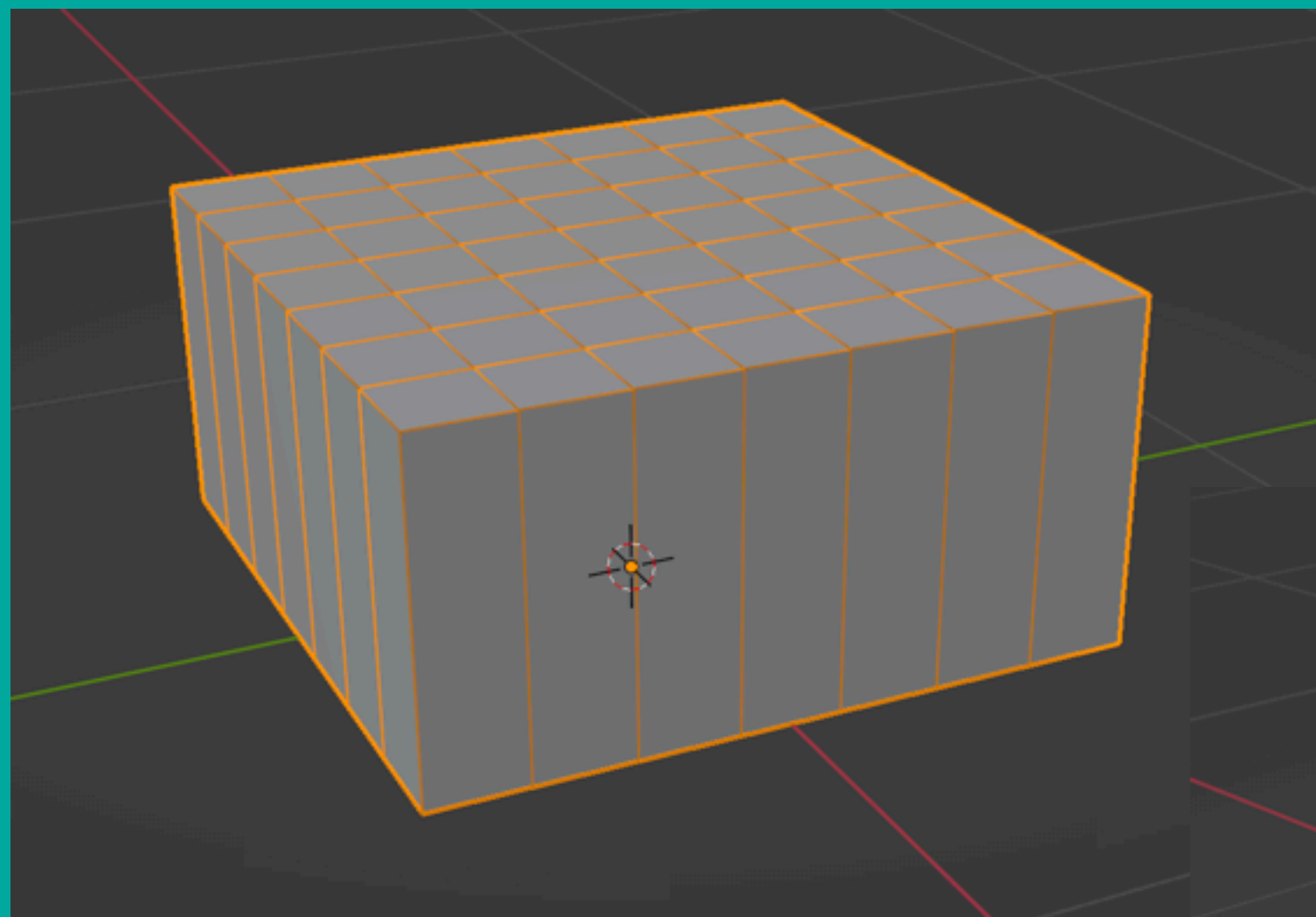
Grid

- Creates a plane
- Set the number of Vertices to get the right grid



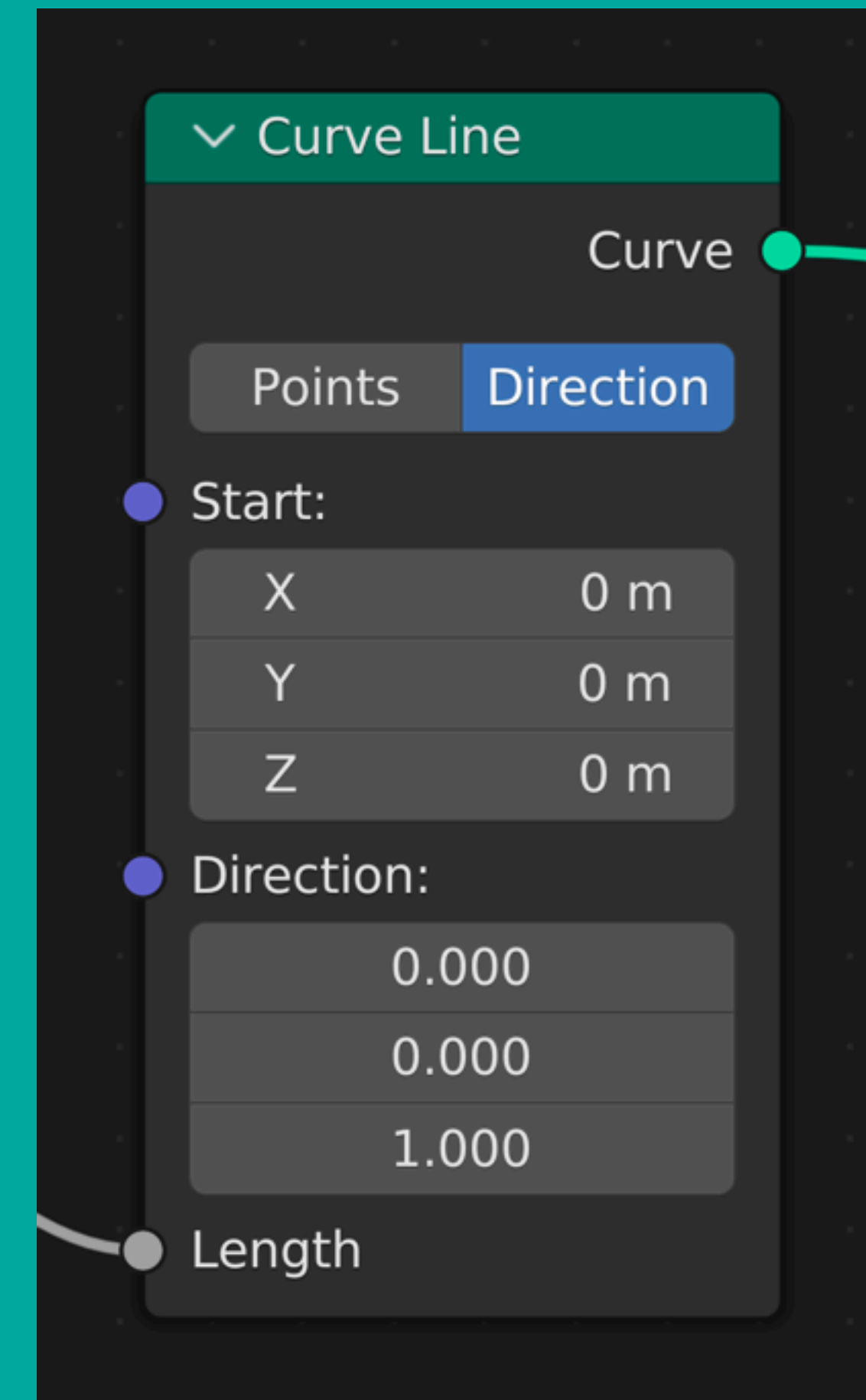
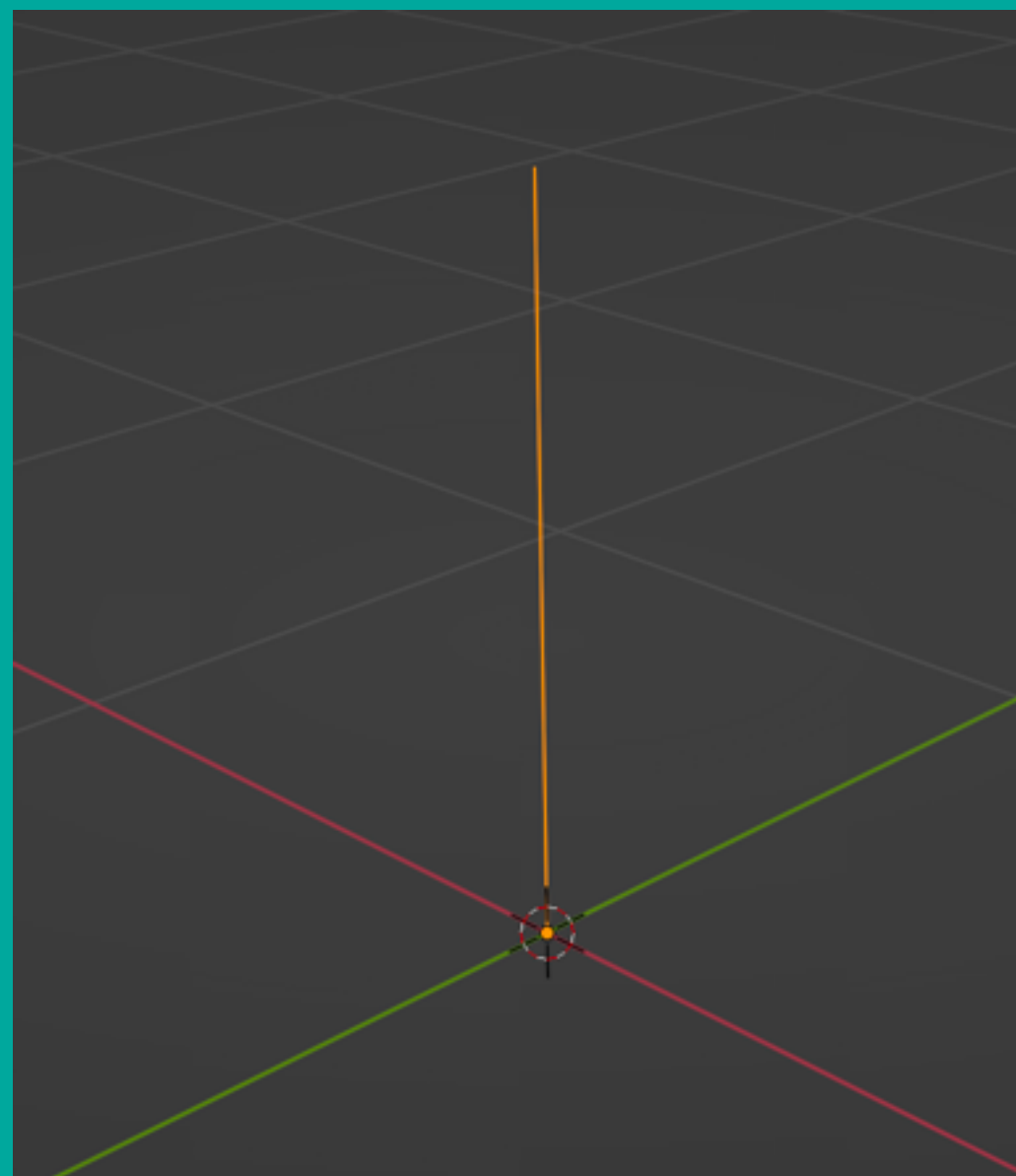
Extrude Mesh

- Extrudes the faces of a mesh (object)
- Use a Vector as Offset input to set the direction of the extrusion



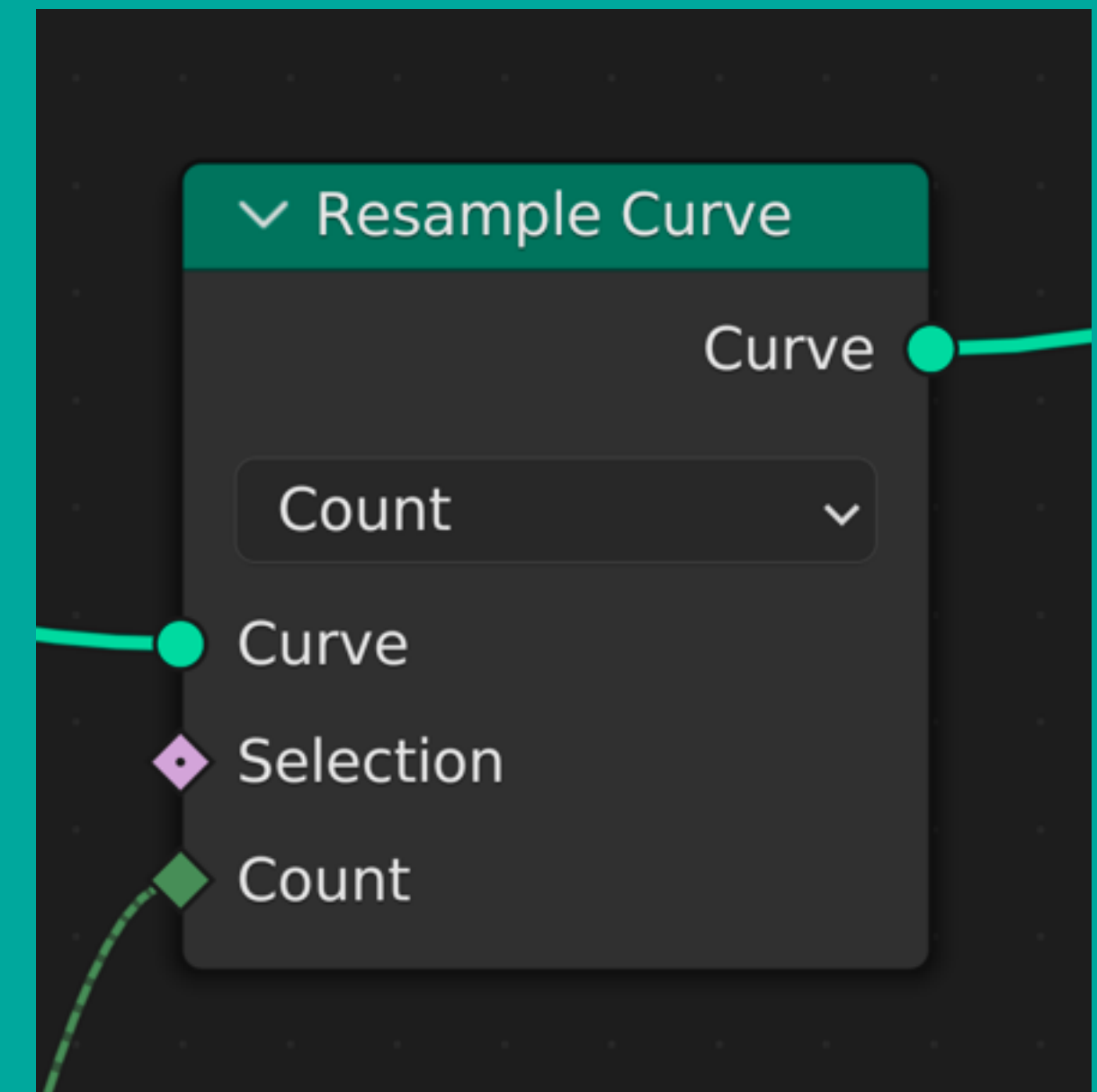
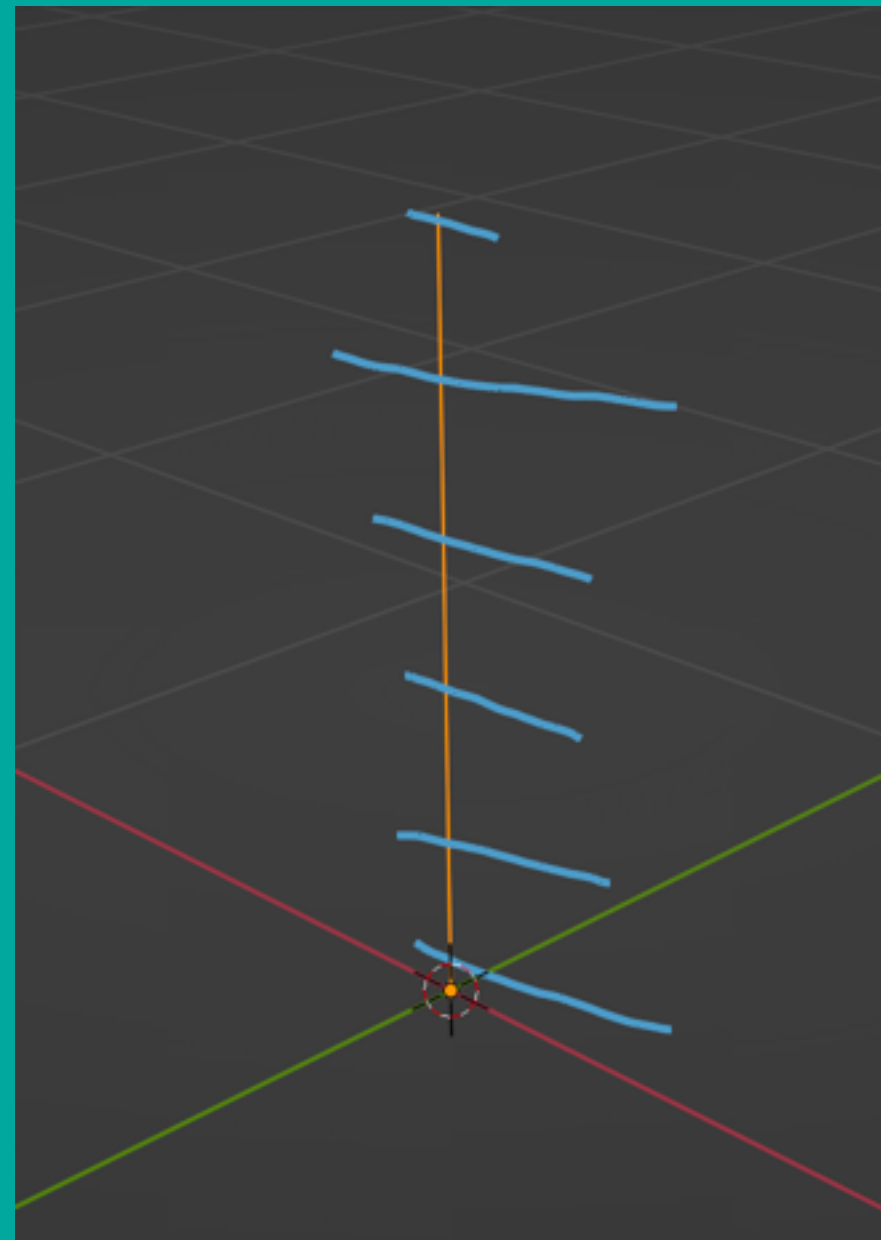
Curve Line

- Generates a line
- Points connects 2 points in space (A to B)
- Direction takes a starting point, a direction, and a length



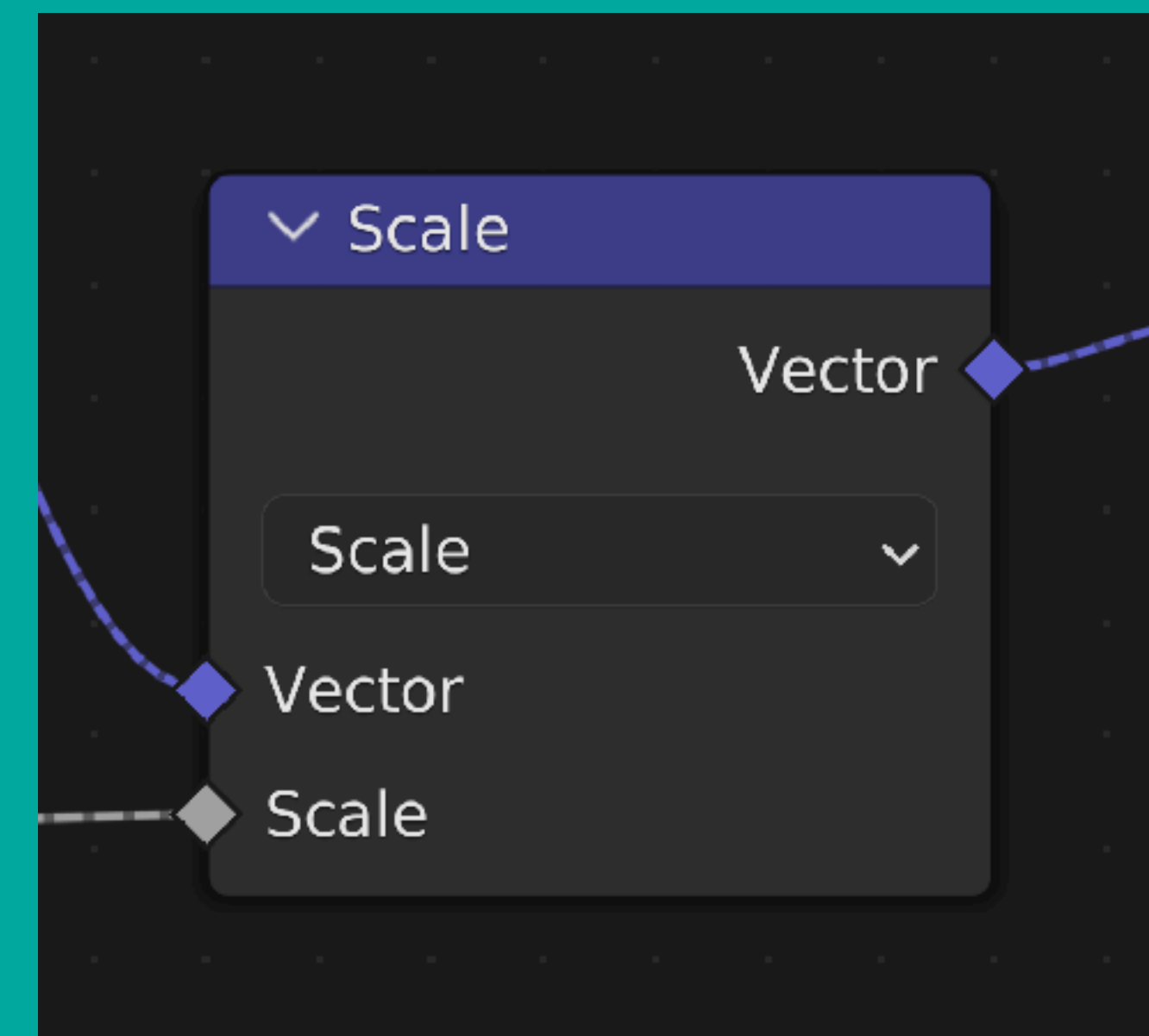
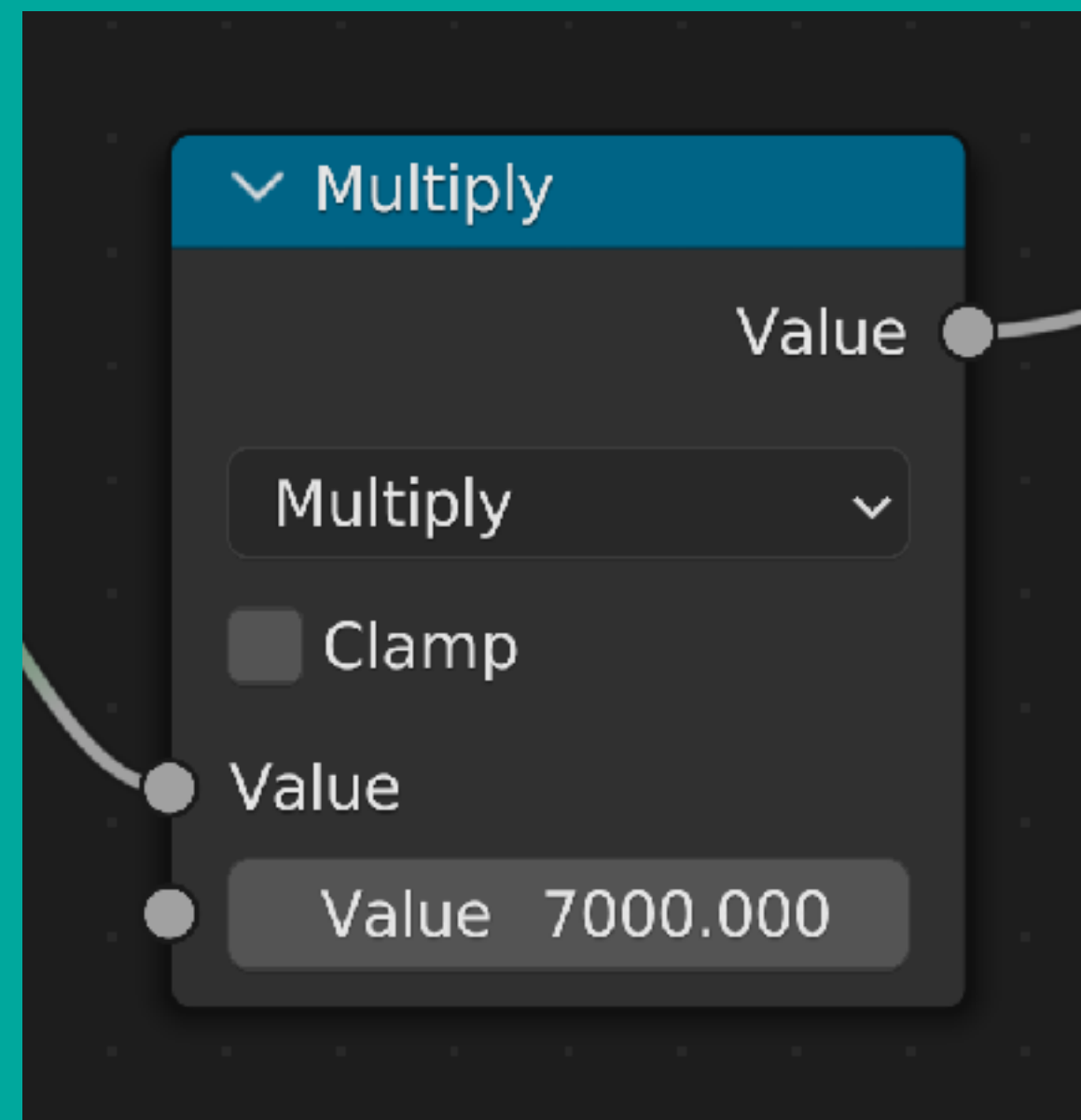
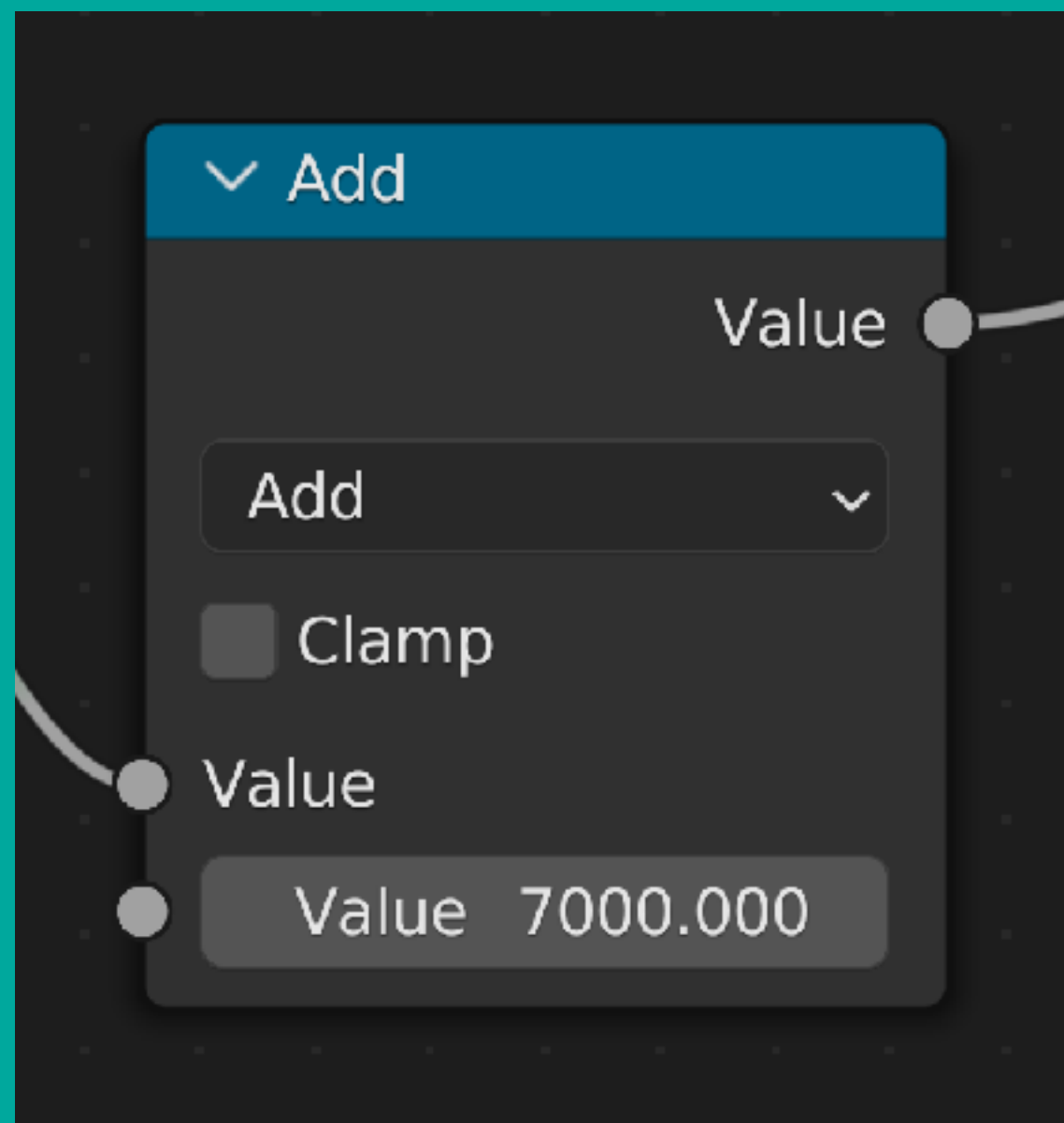
Resample Curve

- Divides the line in segments
- Use count to set number of segments



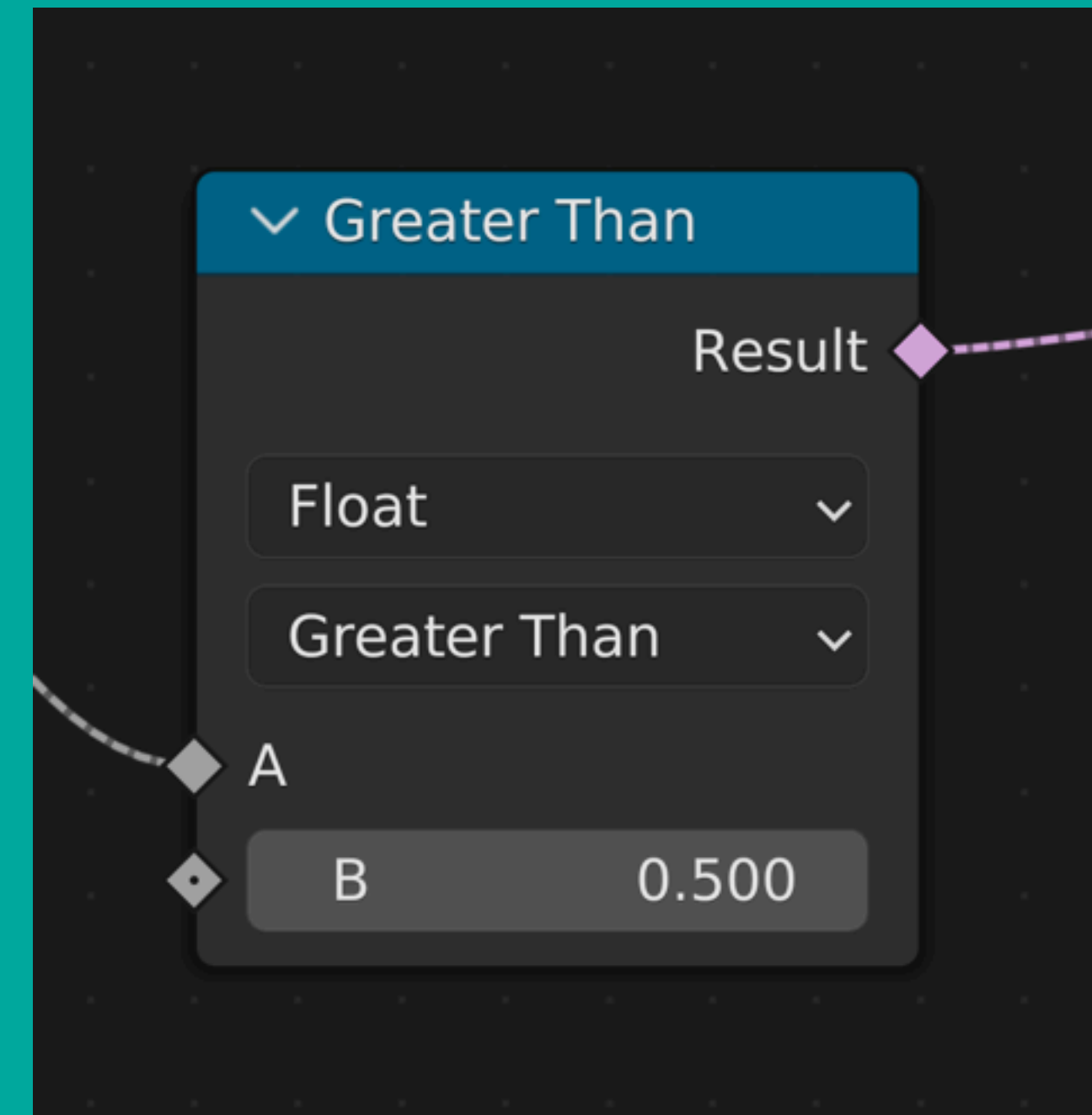
Math

- Can do all kinds of math functions!
- Add (+), subtract (−), divide (/), multiply (*) and much more



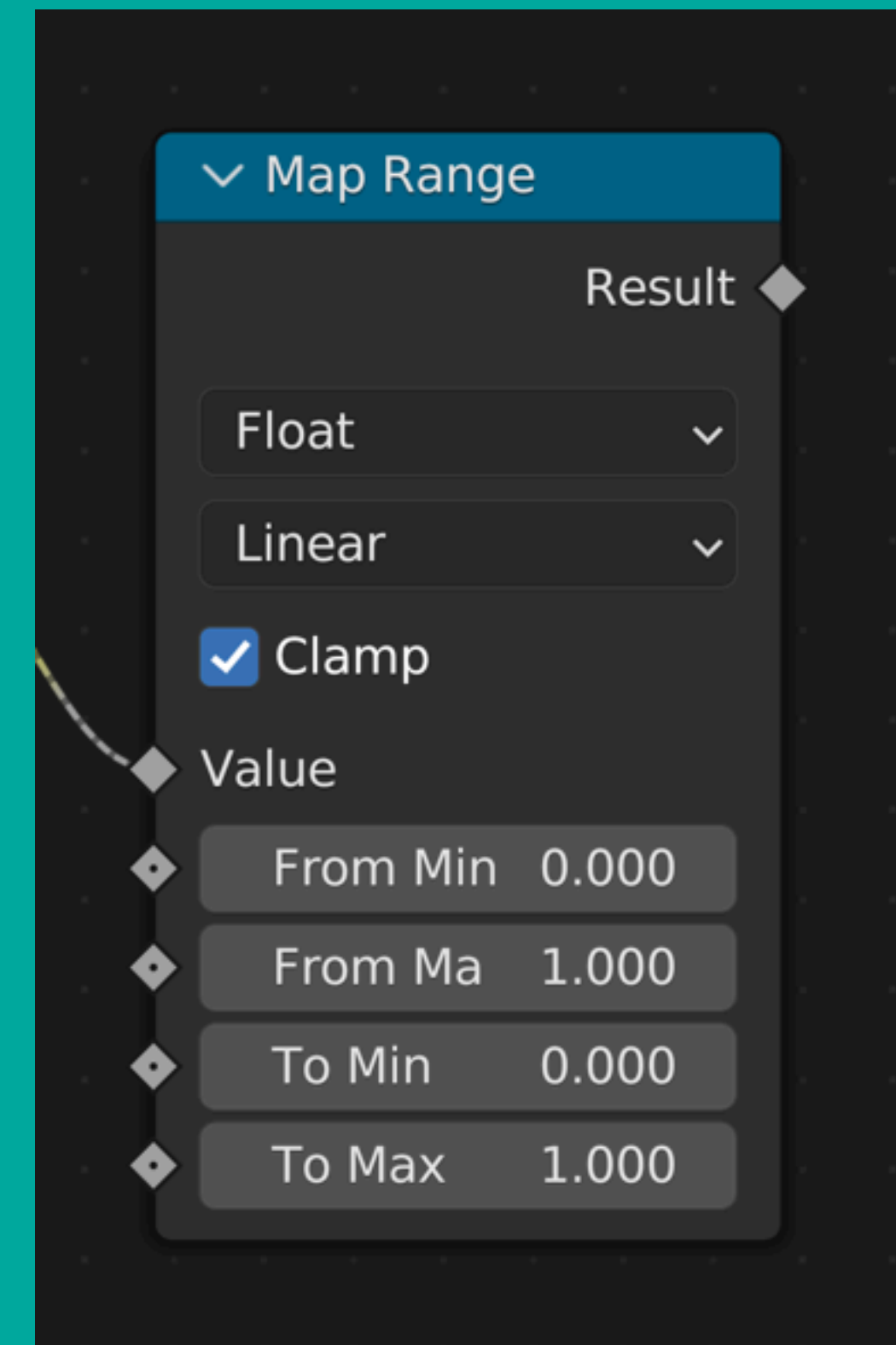
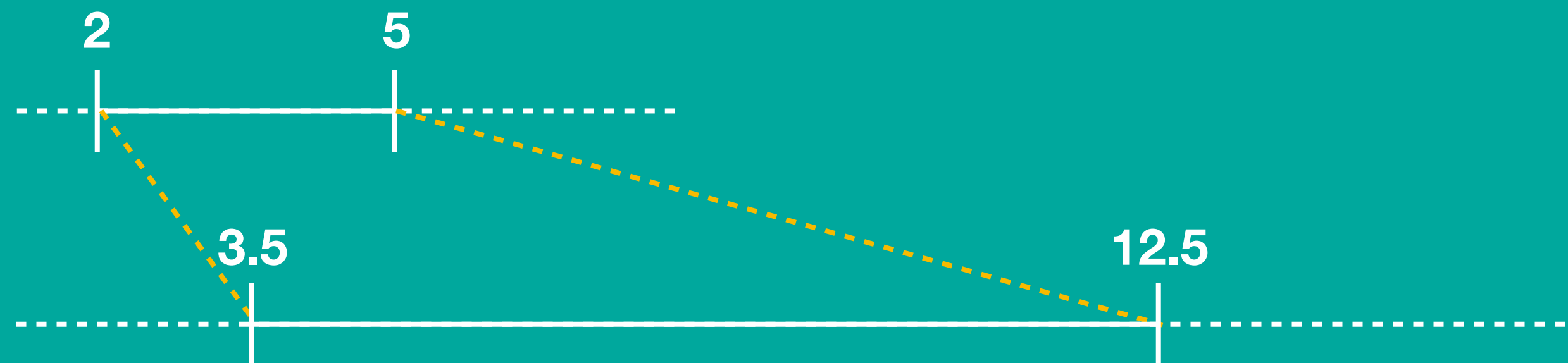
Compare

- Compares the input value to another value
- Like when you would do $A > B$ in math
- When connecting a random value between 0 and 1, you can do something with half of the selected elements



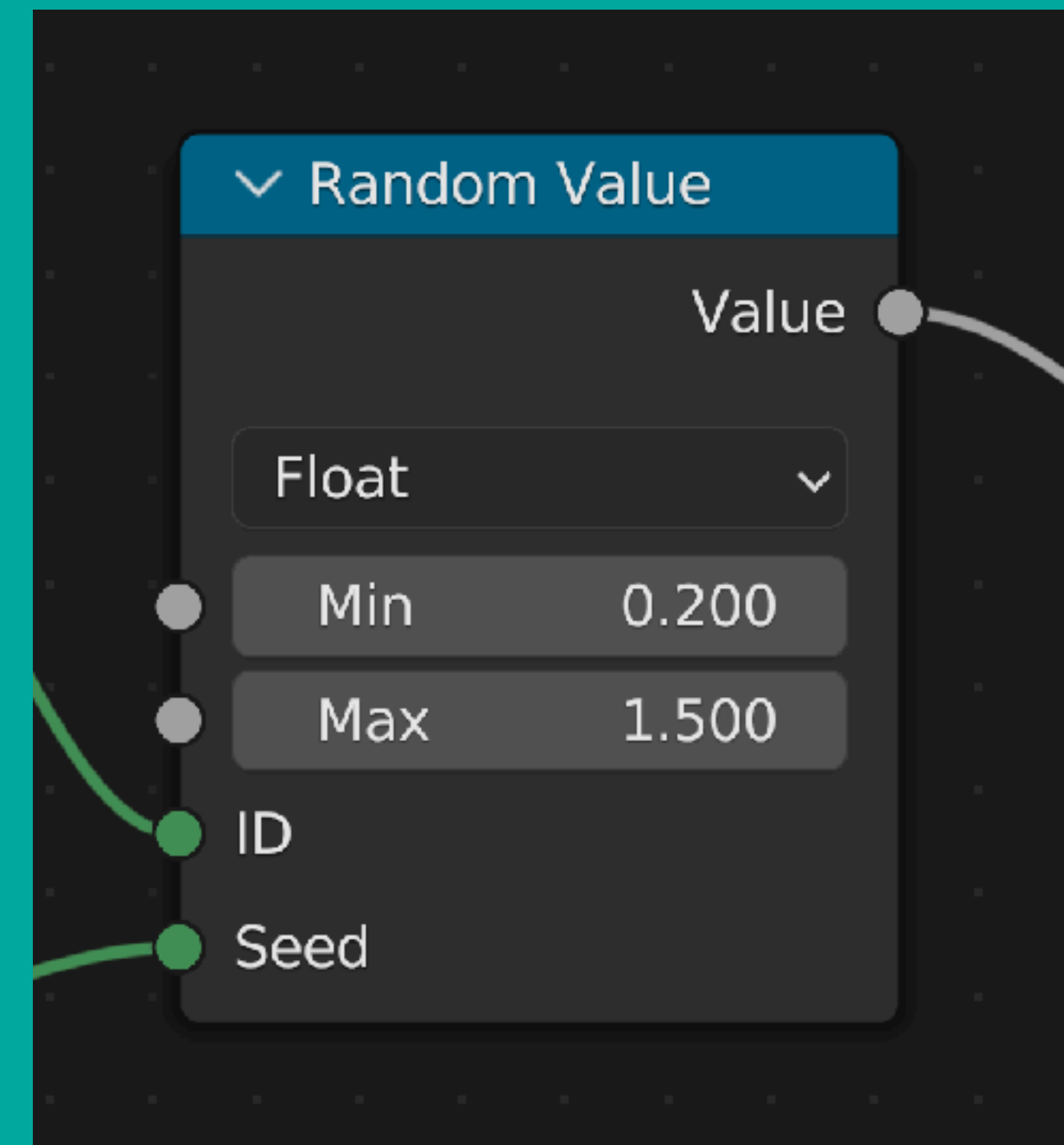
Map Range

- Translates a range to another range
- Use it for instance to translate a random number between 0 and 1 to -1 and 1



Random Value

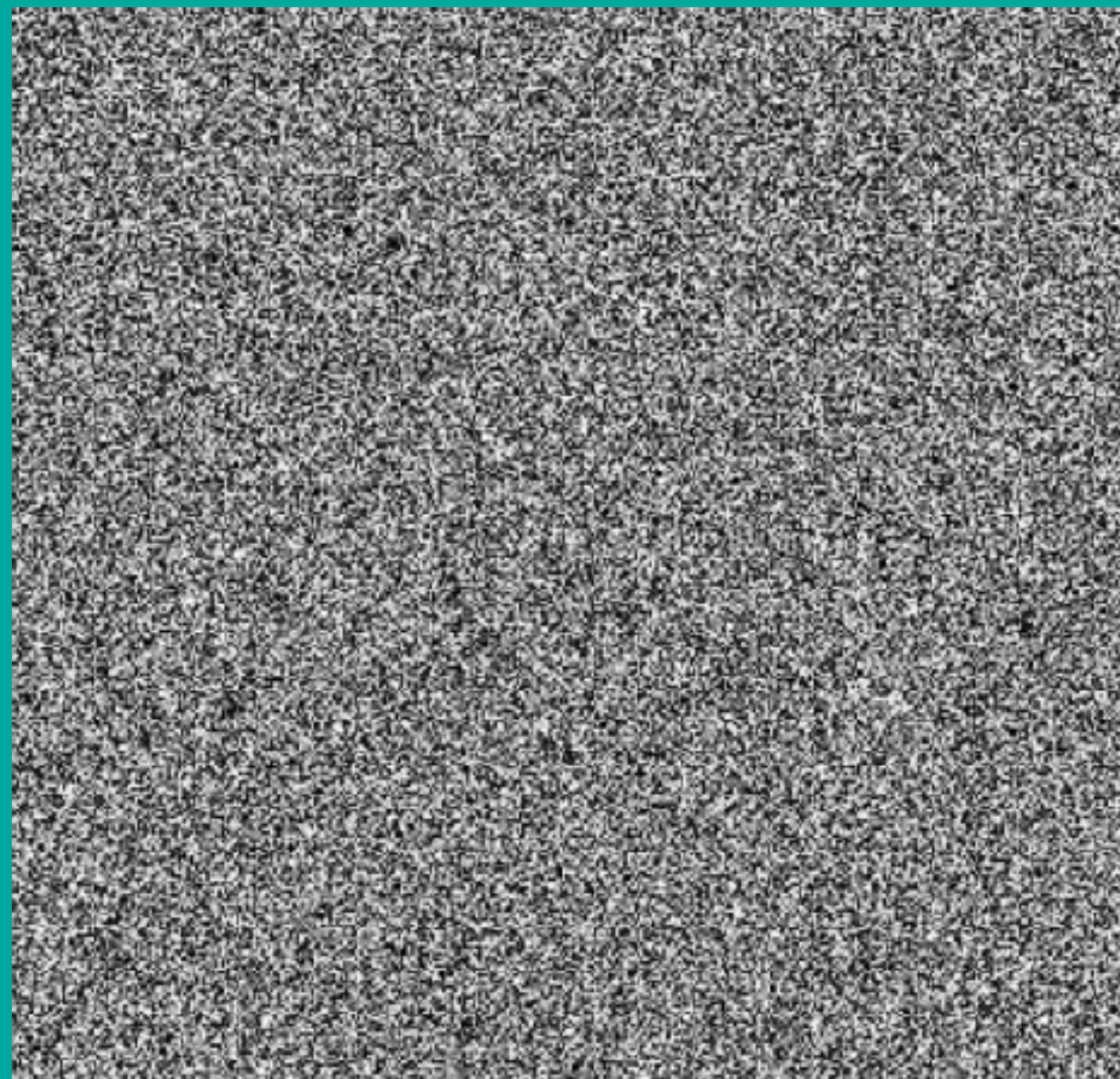
- Generates a random value between Min and Max
- Seed defines the set of random numbers it will generate
- ID defines the index of the set of random numbers it will generate (so the 1st, the 2nd, or the nth)
- Given the same Seed and the same ID, you will always get the same random value!



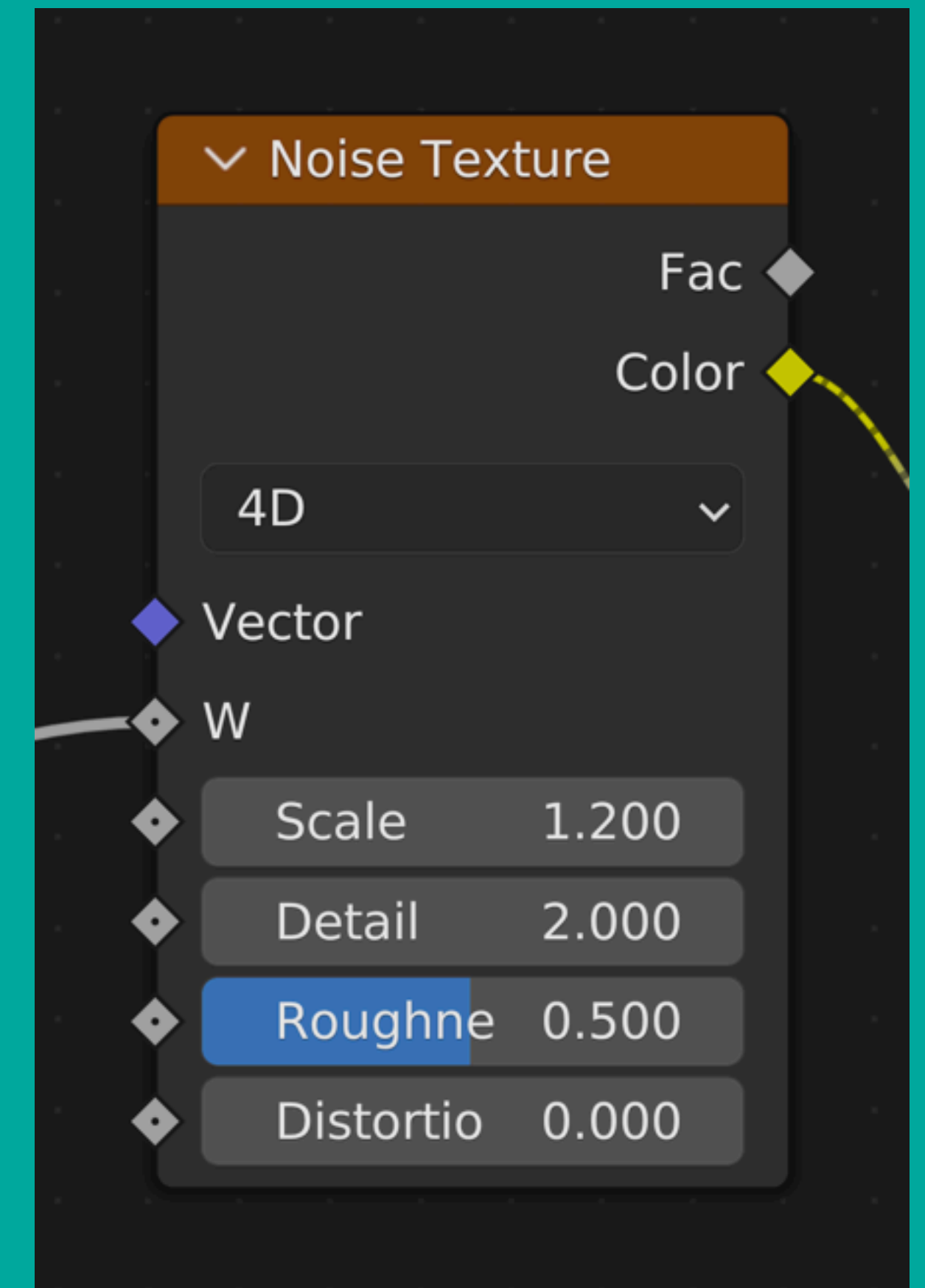
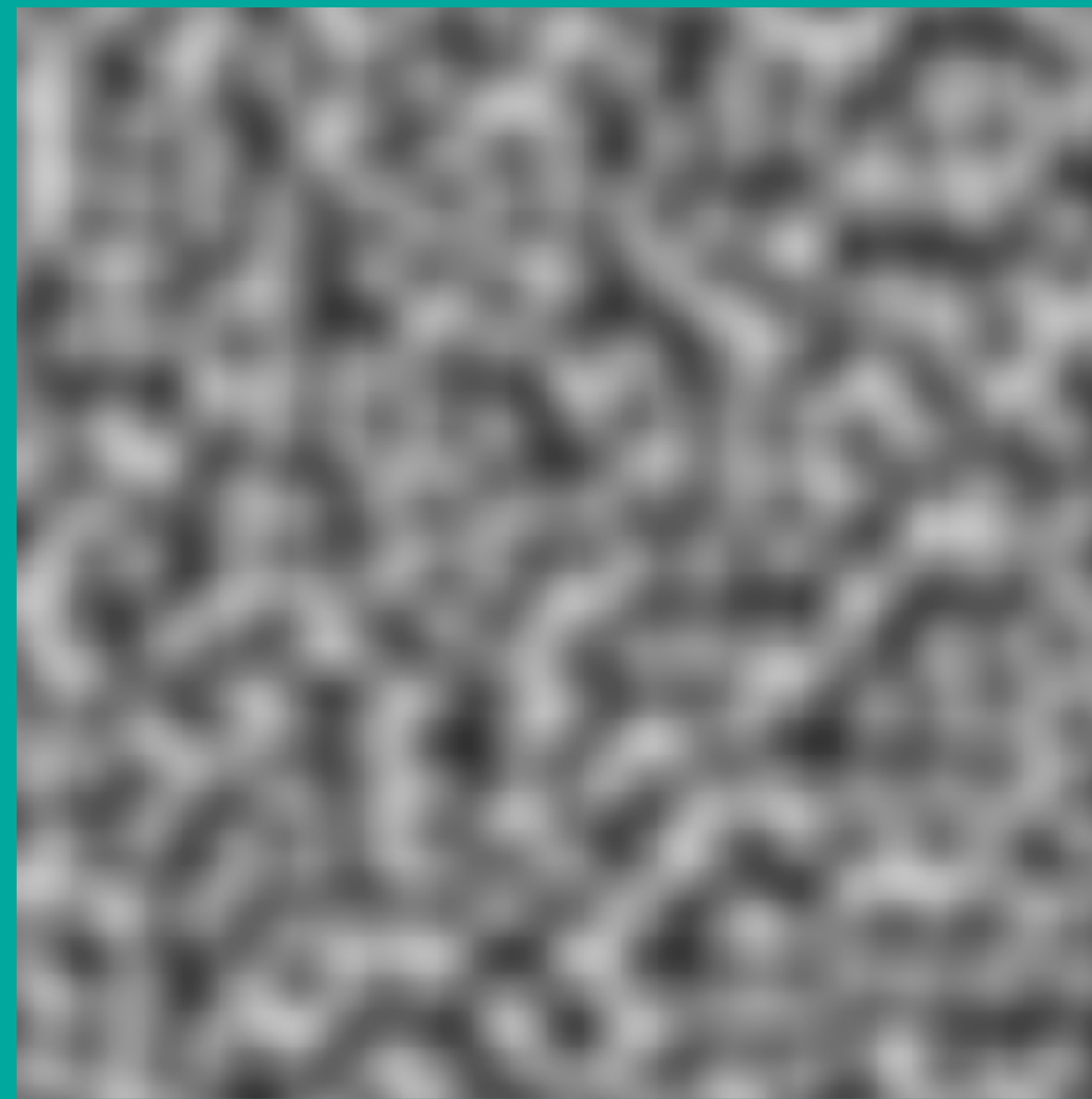
Noise Texture

- Generates smooth random values
- Based on Perlin noise texture

Random

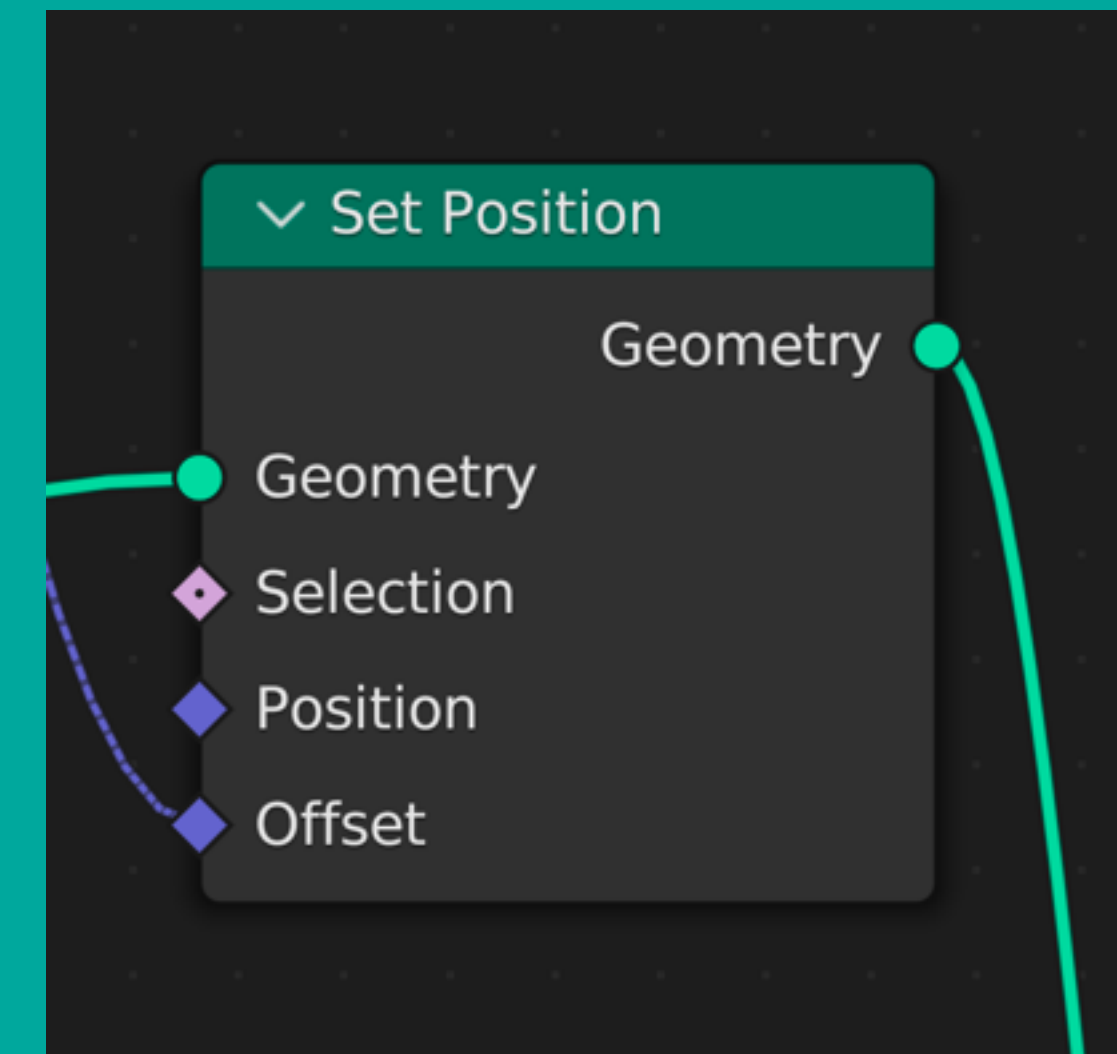


Perlin



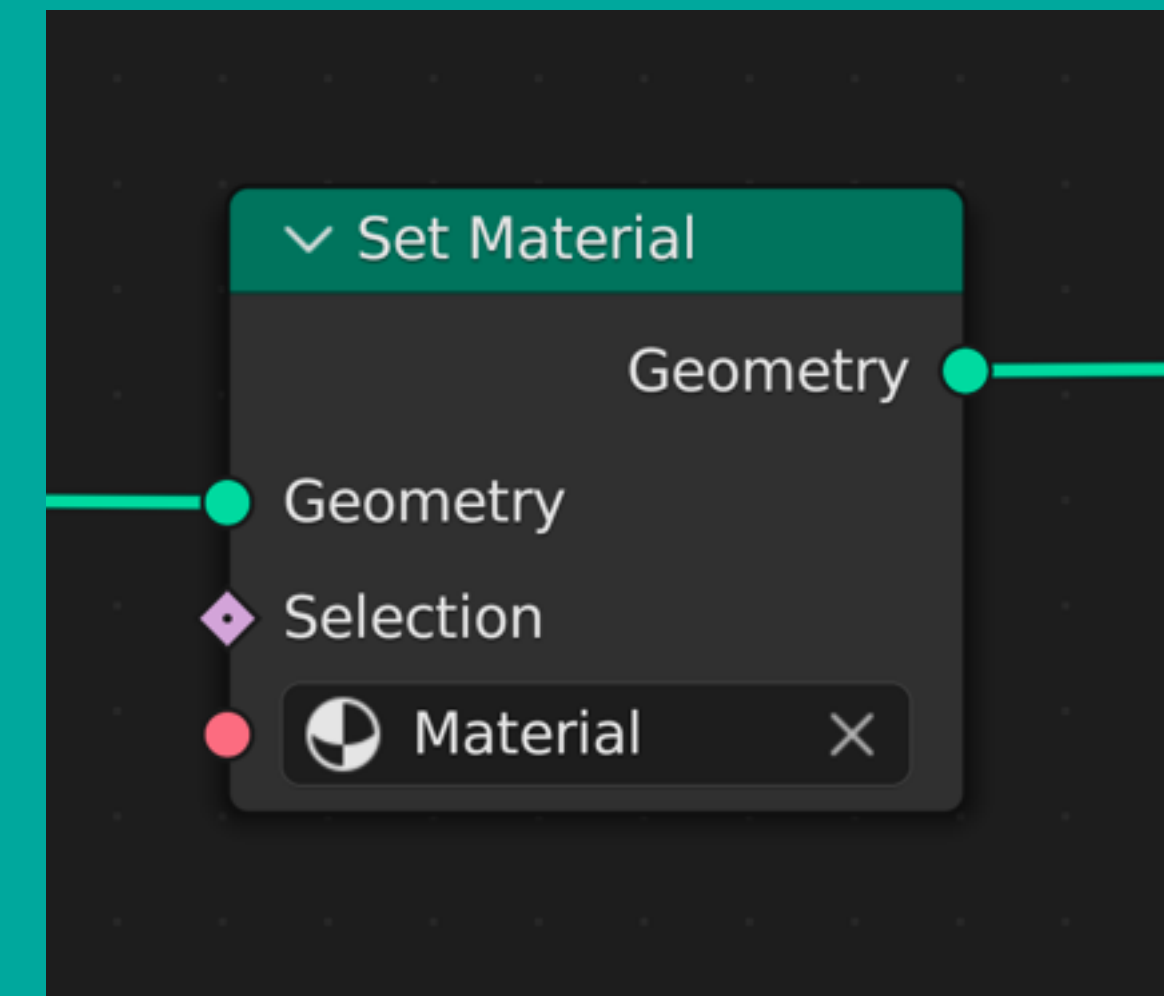
Set Position

- Moves things
- Use **position** to move something to a specific position; 'Move the chair to the corner of the room'
- Use **offset** to move something a specific distance; 'Move the chair two meters to the left and one meter forwards'



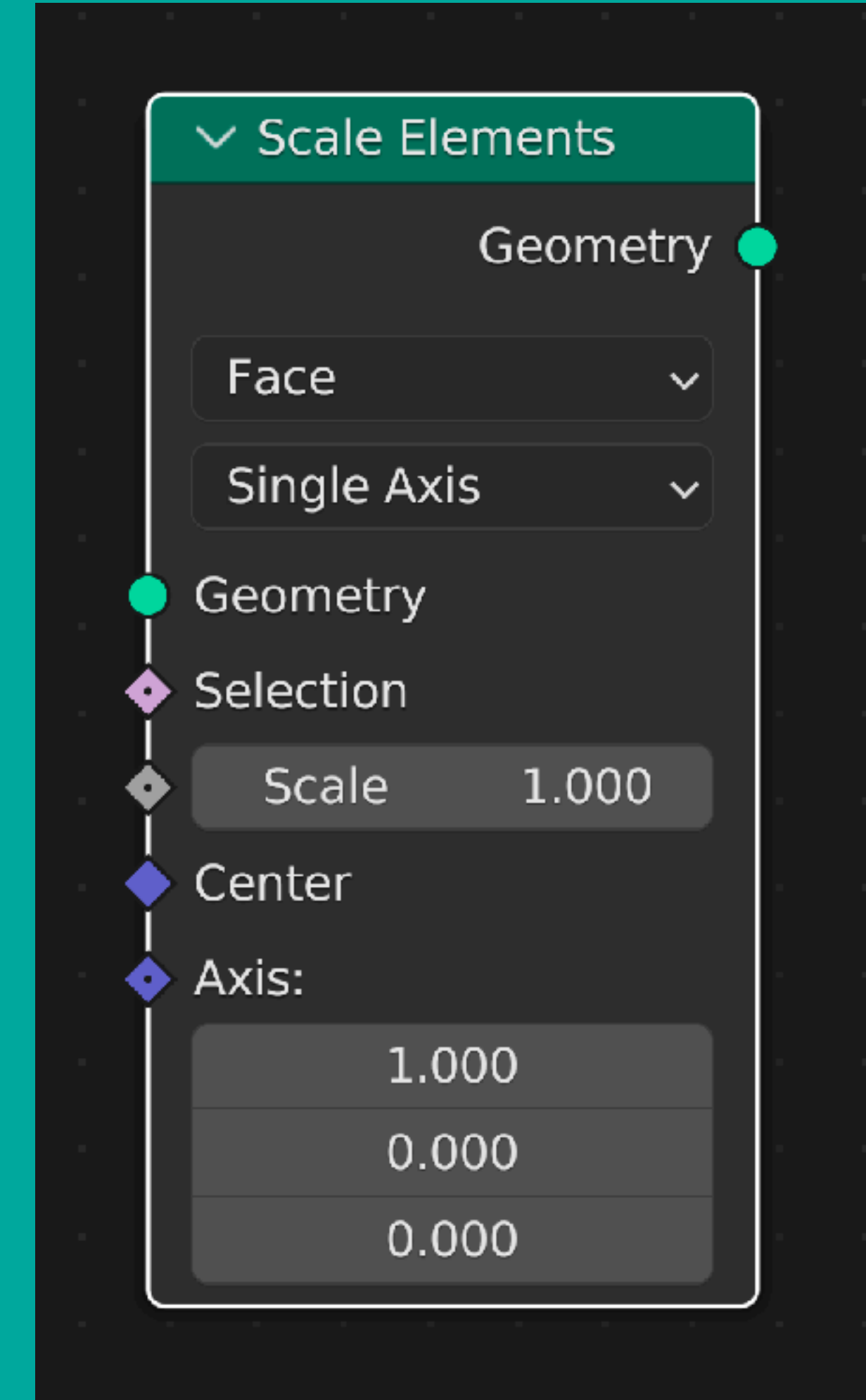
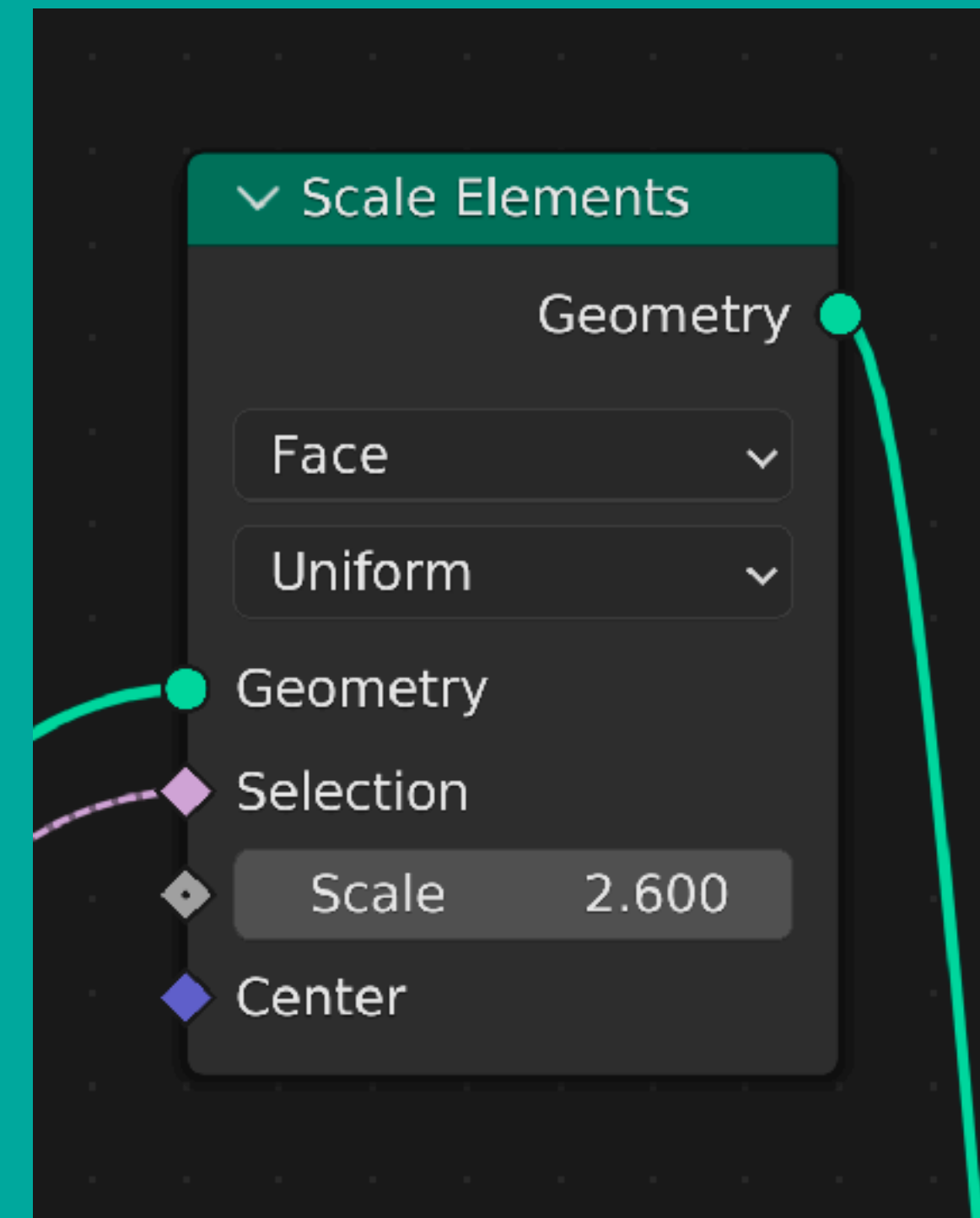
Set Material

- Sets material on Geometry
- You have to create the material first in the Material Properties panel



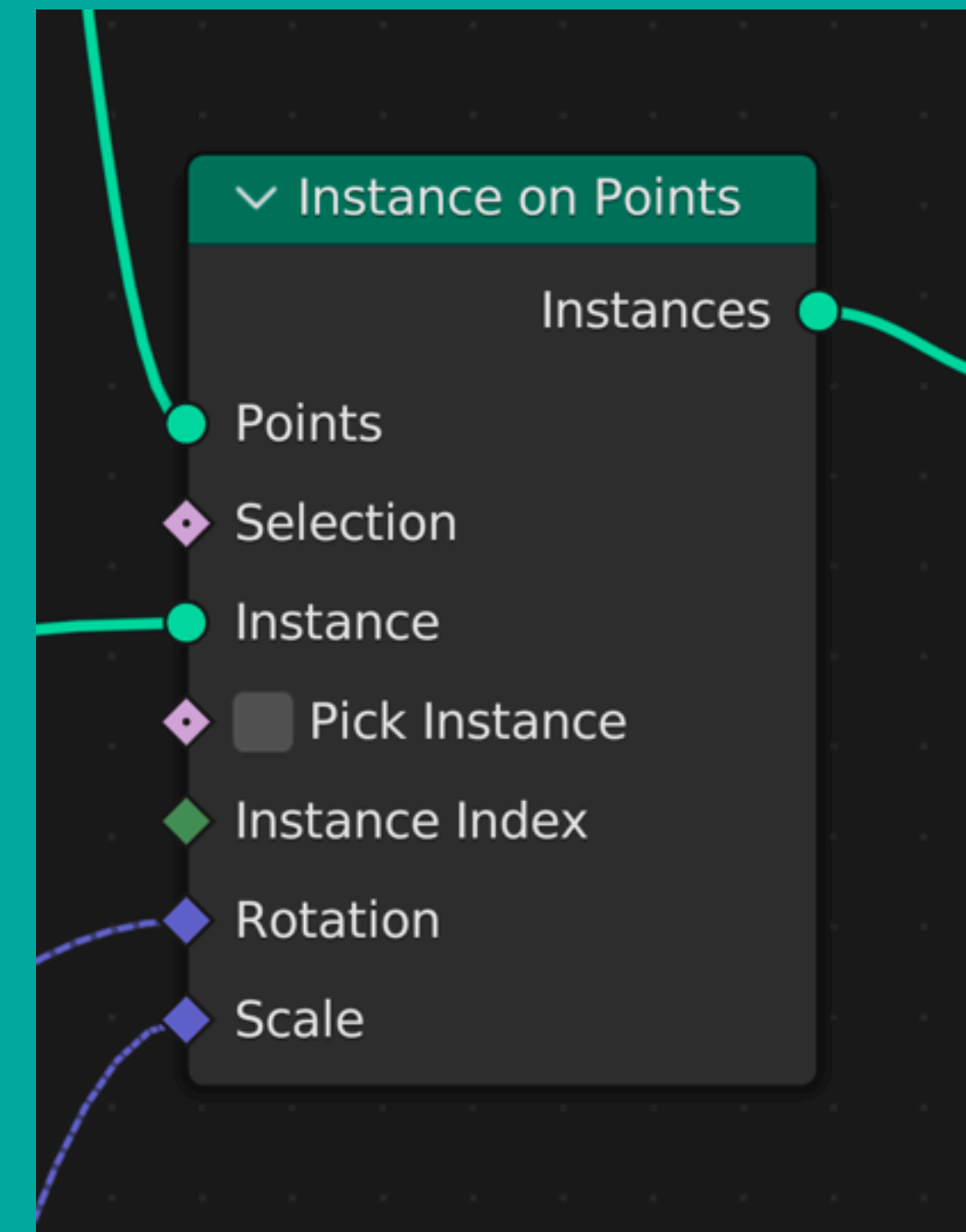
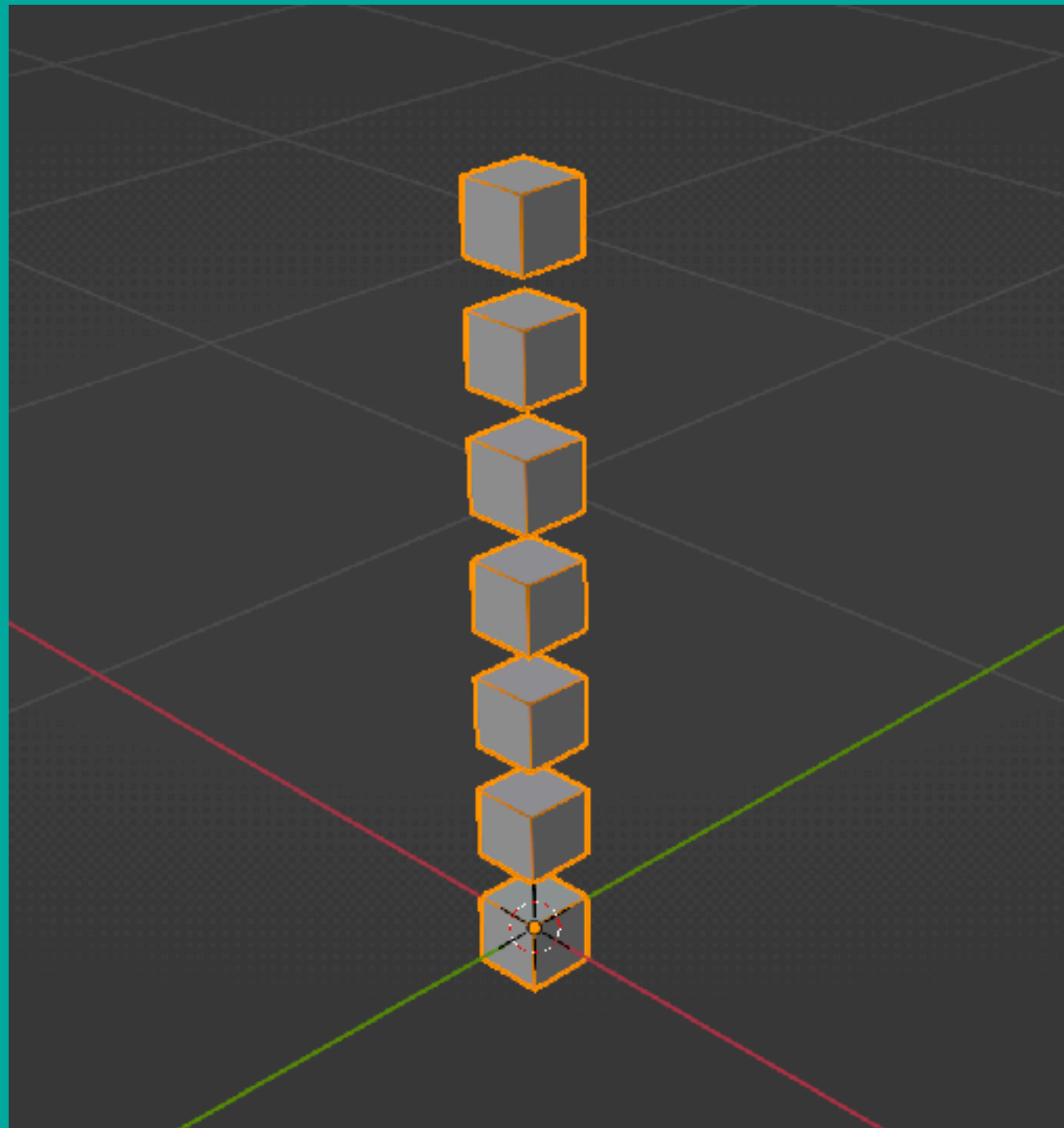
Scale Elements

- Scales elements, so it makes them bigger or smaller
- Change 'Uniform' to 'Single Axis' to scale non-proportionally



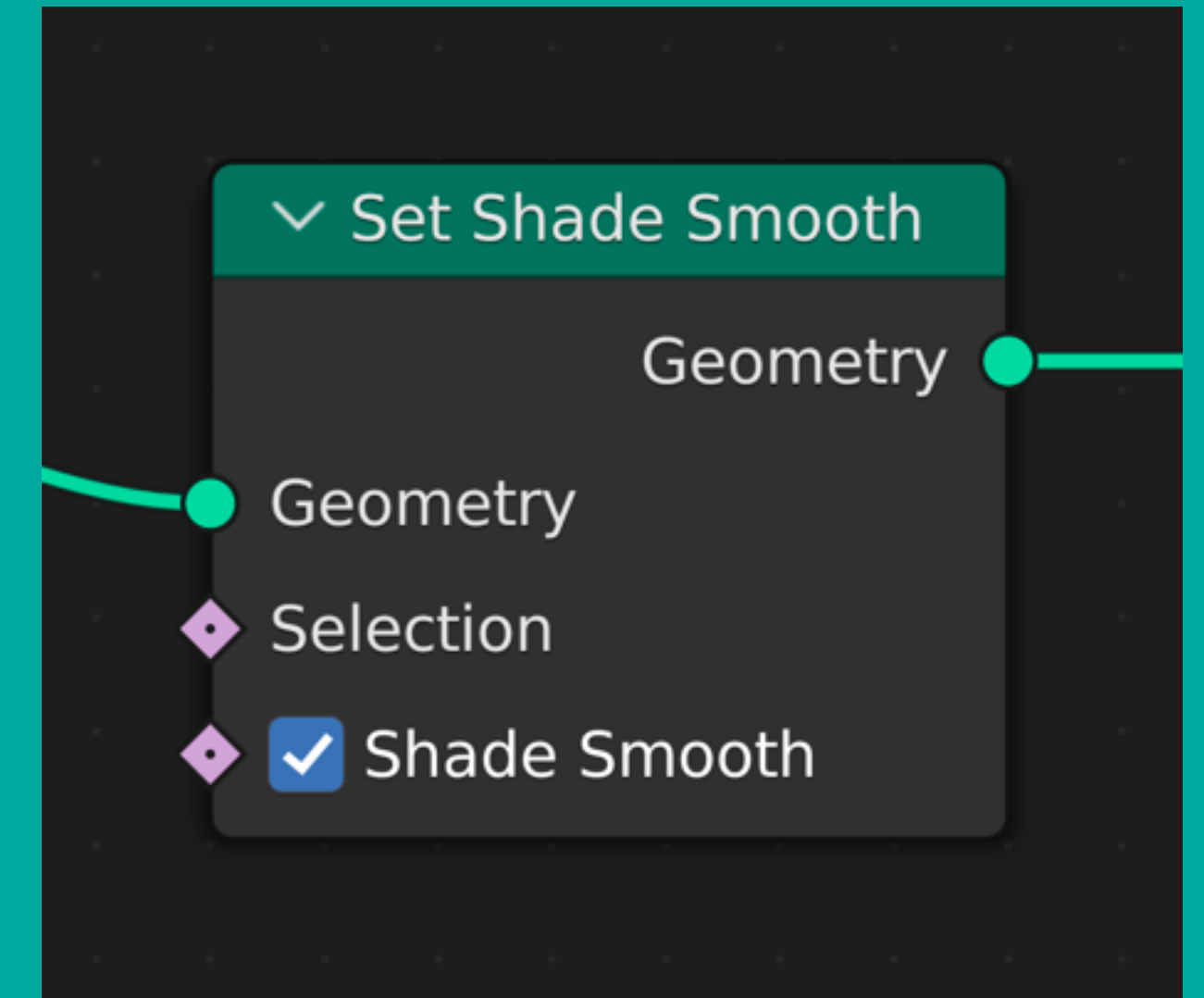
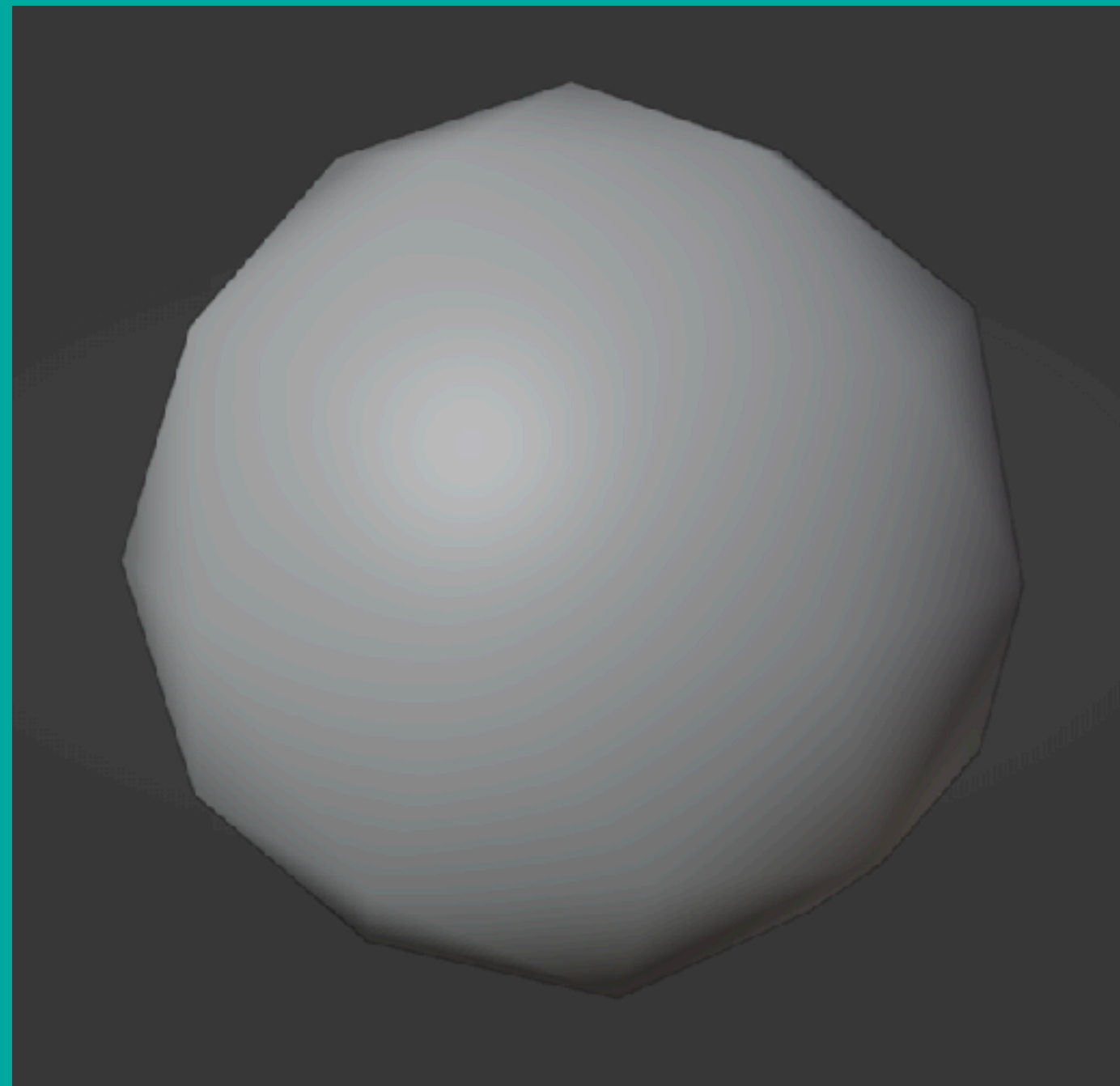
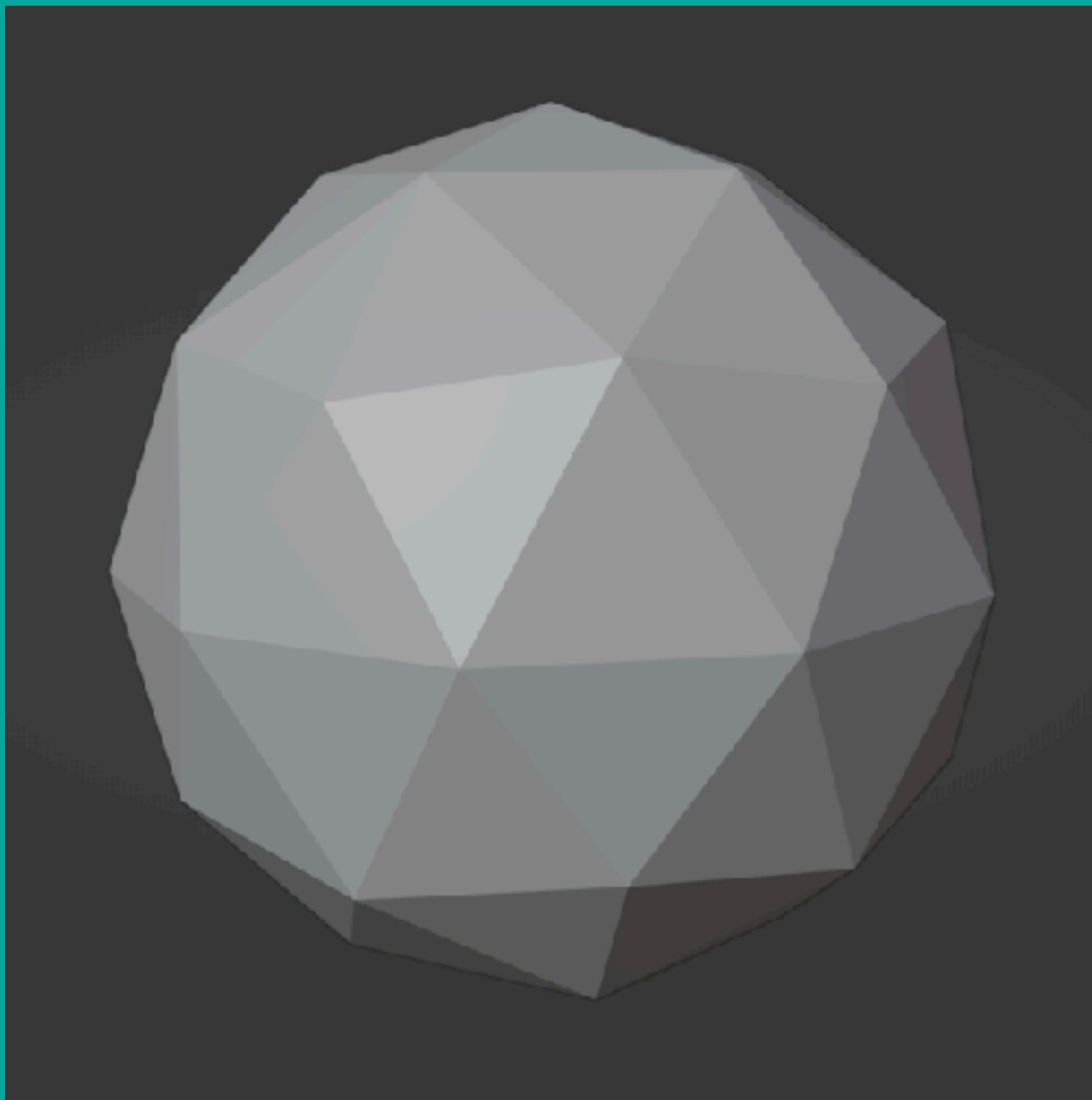
Instance on Points

- Copies instances (objects) on points
- Set the rotation and size of each of the points



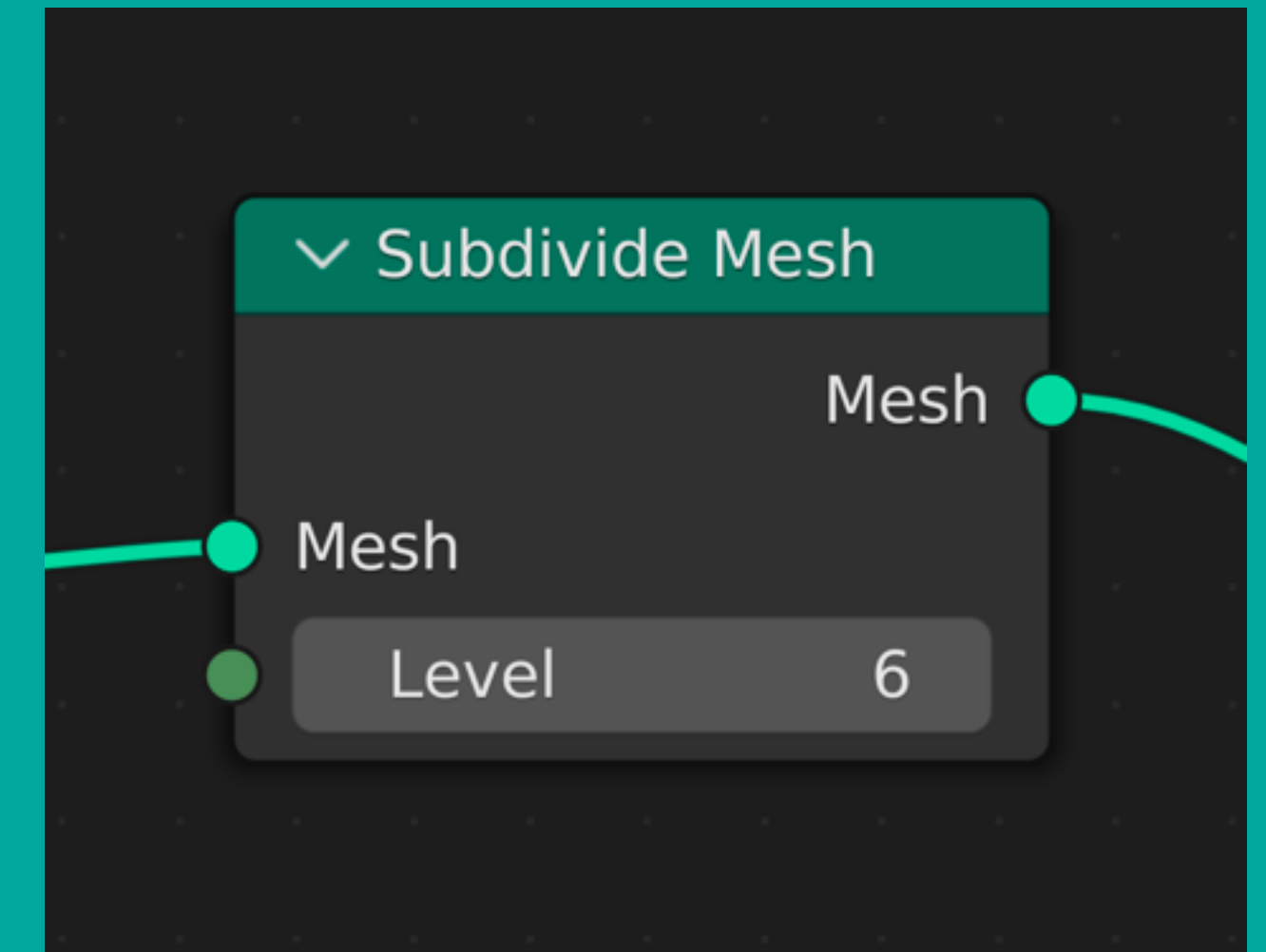
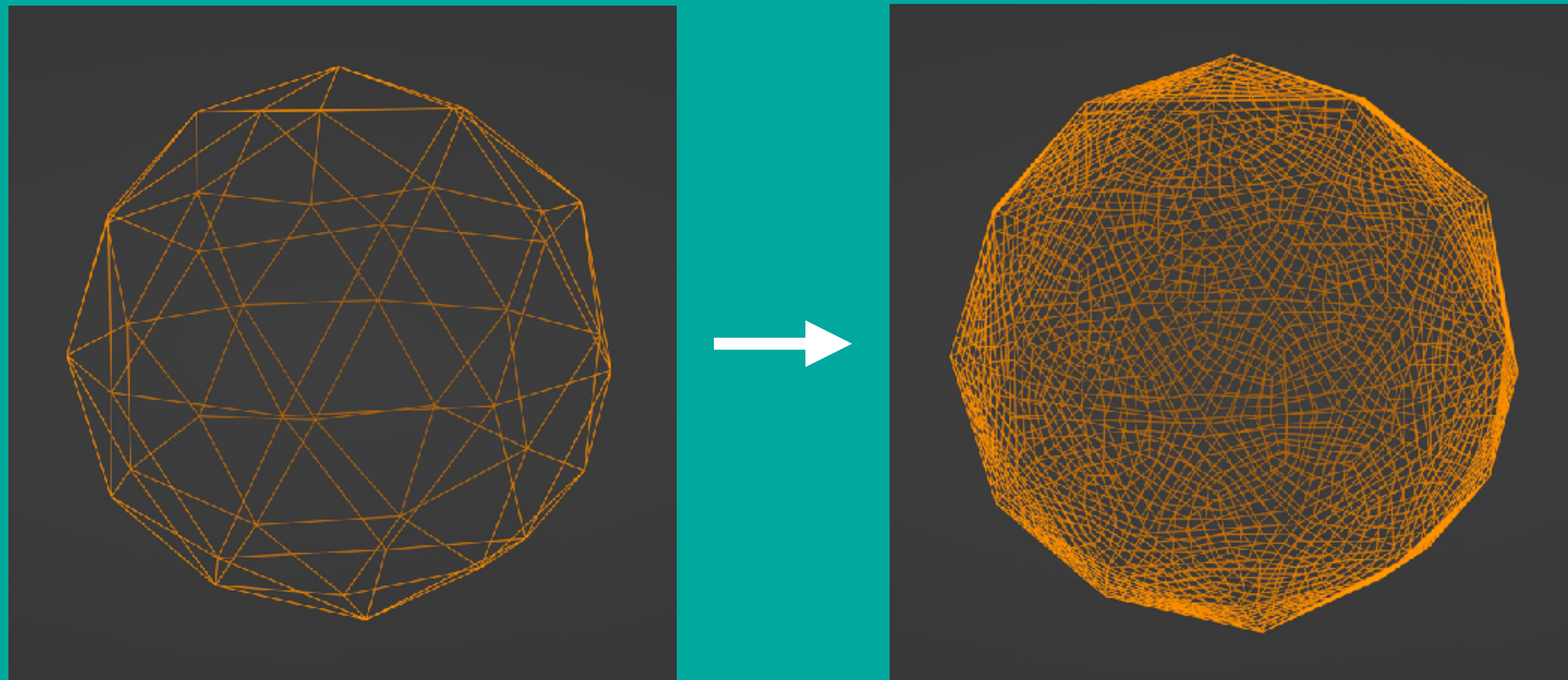
Set Shade Smooth

- Smooths a mesh



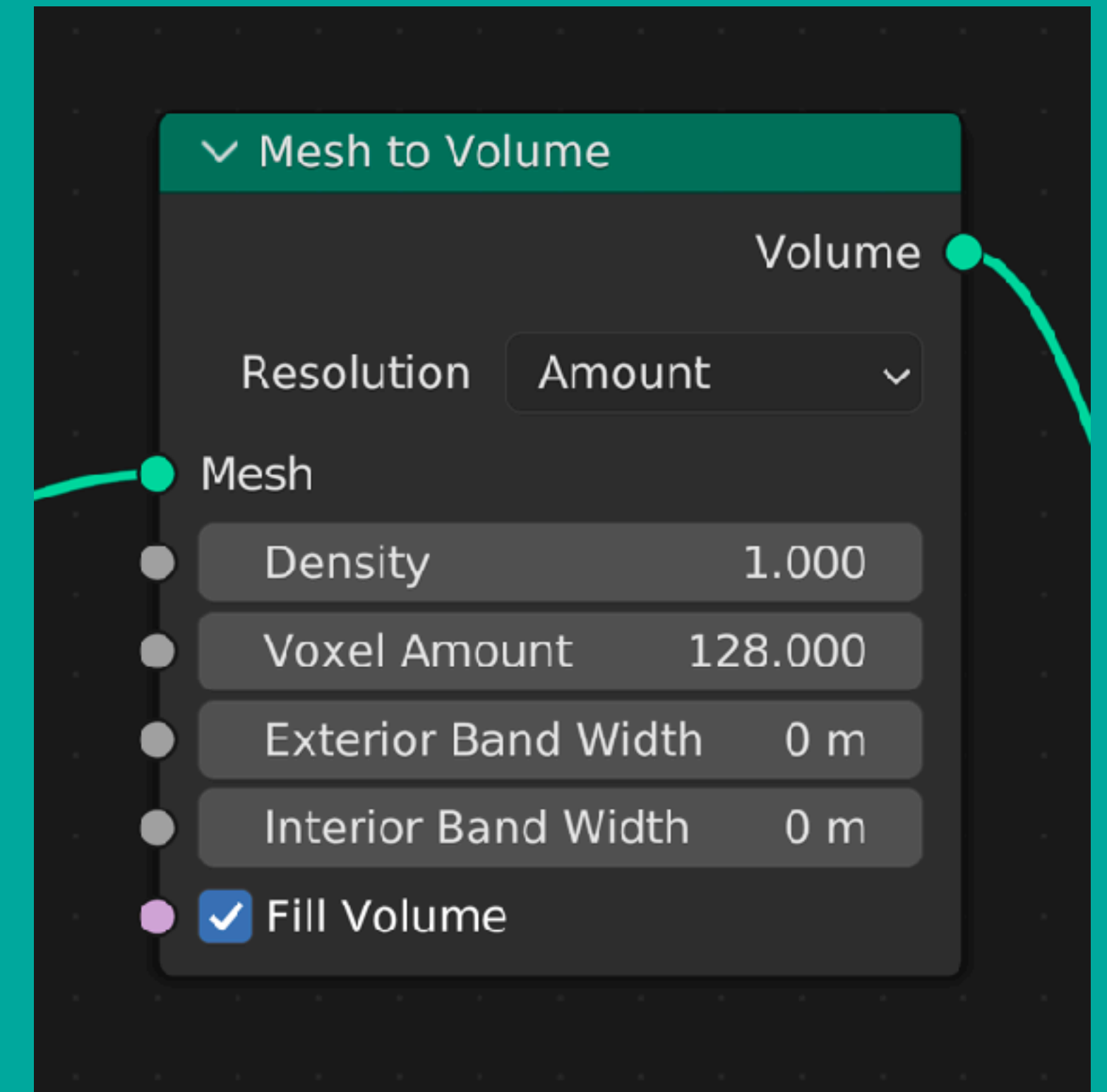
Subdivide Mesh

- Divides a mesh to add more detail by adding subdivisions



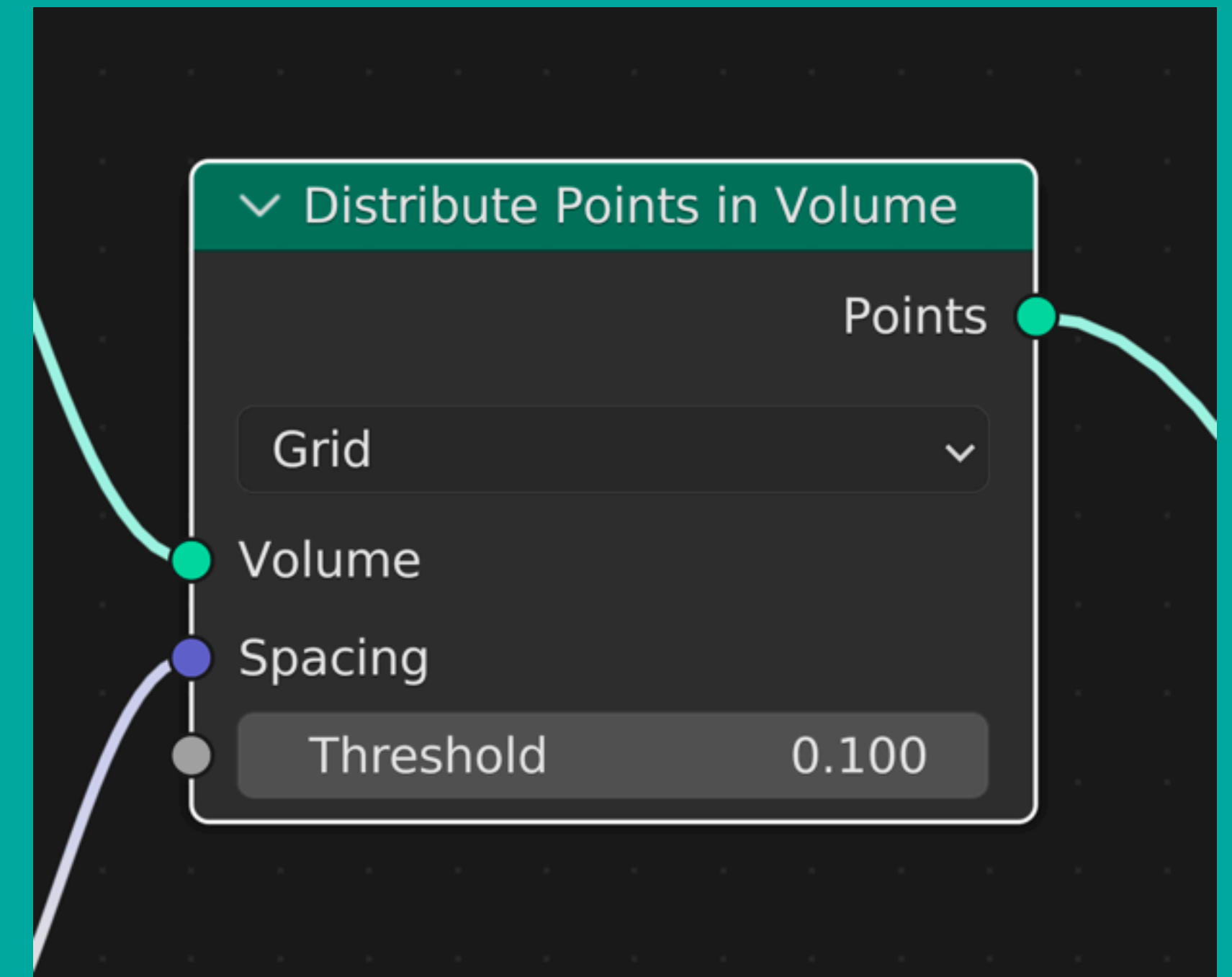
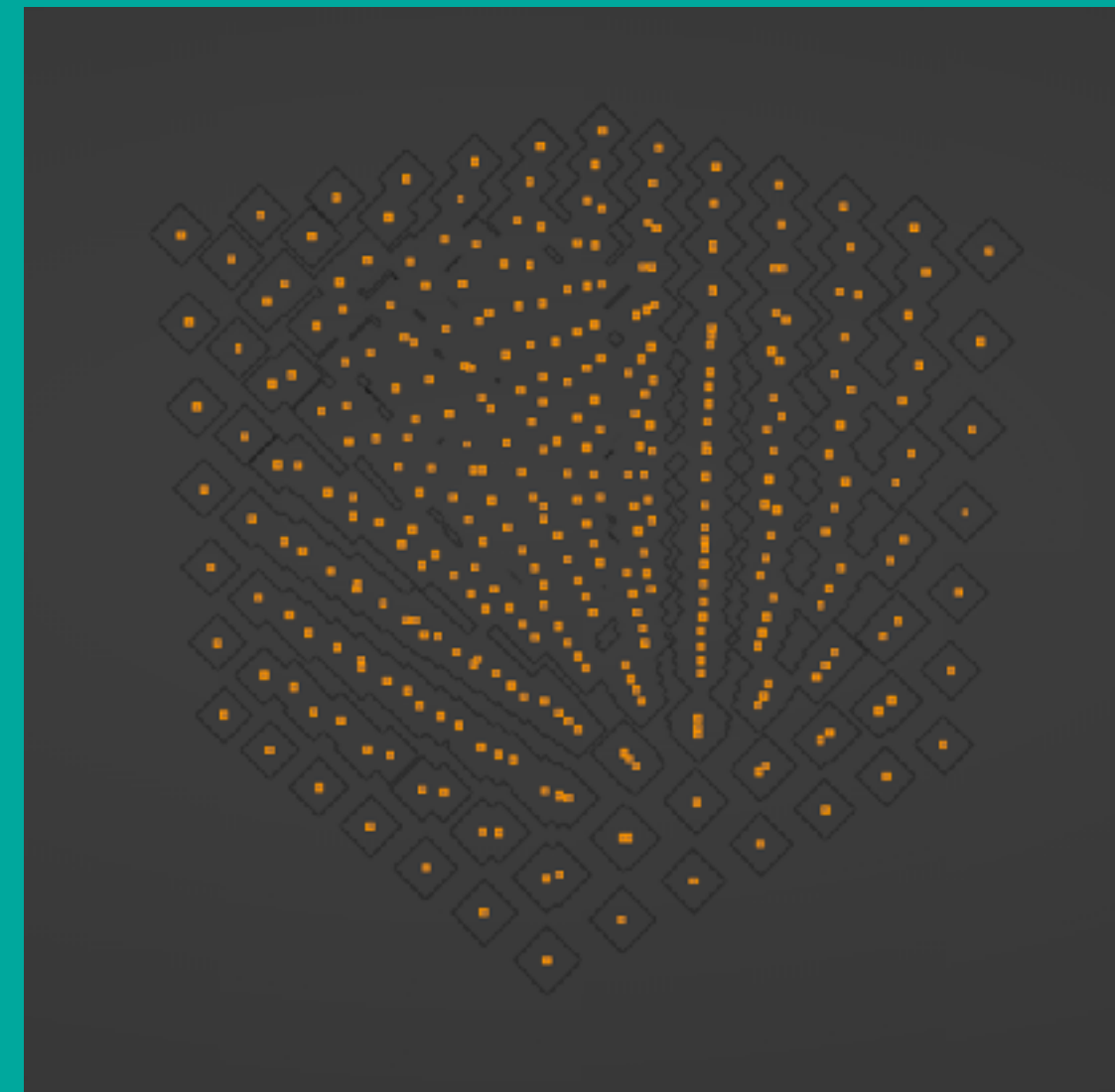
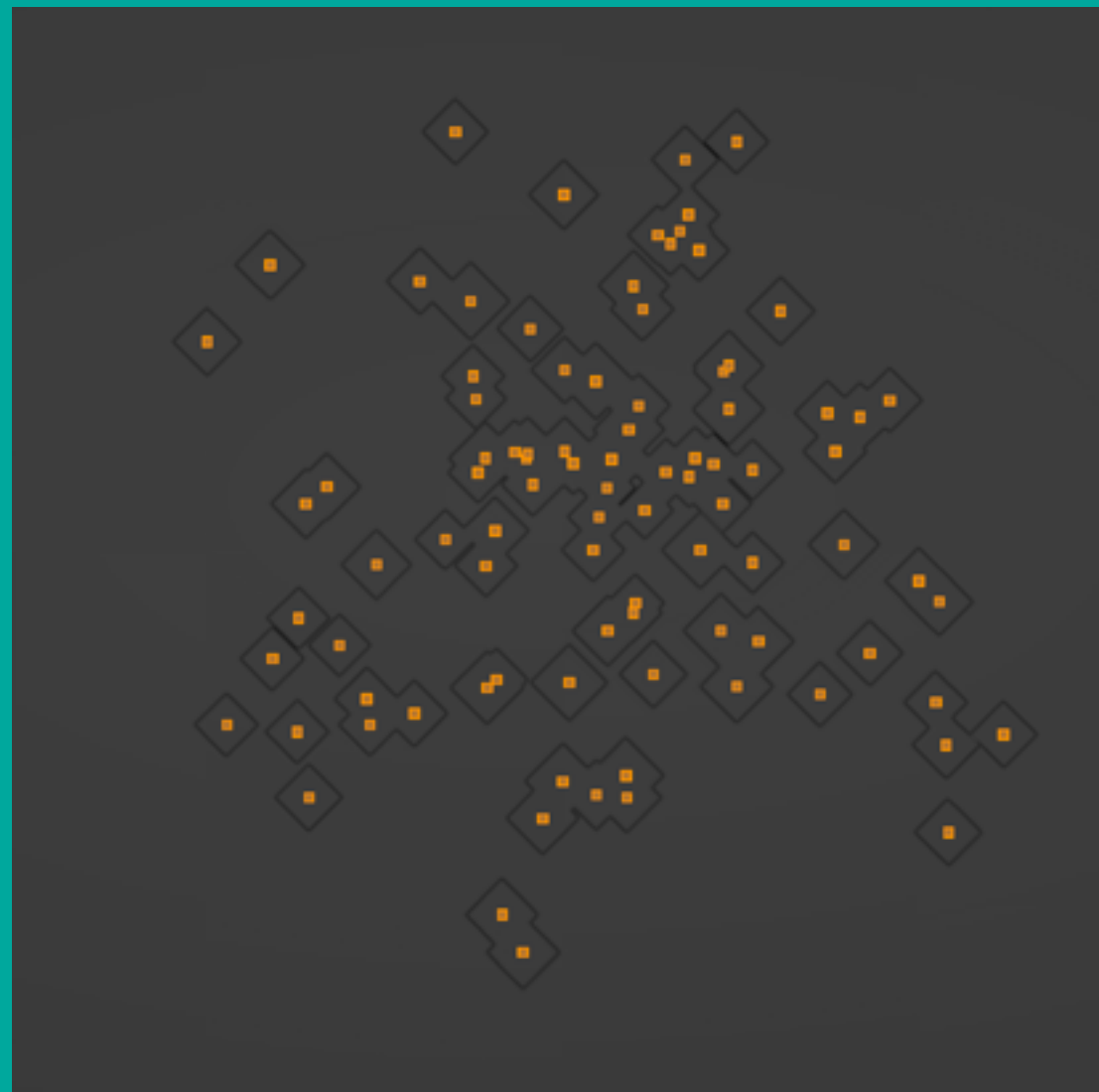
Mesh to Volume

- Converts an empty object (surface) to a filled object (the inside)
- Set exterior and interior Band Width to 0 to keep your original shape



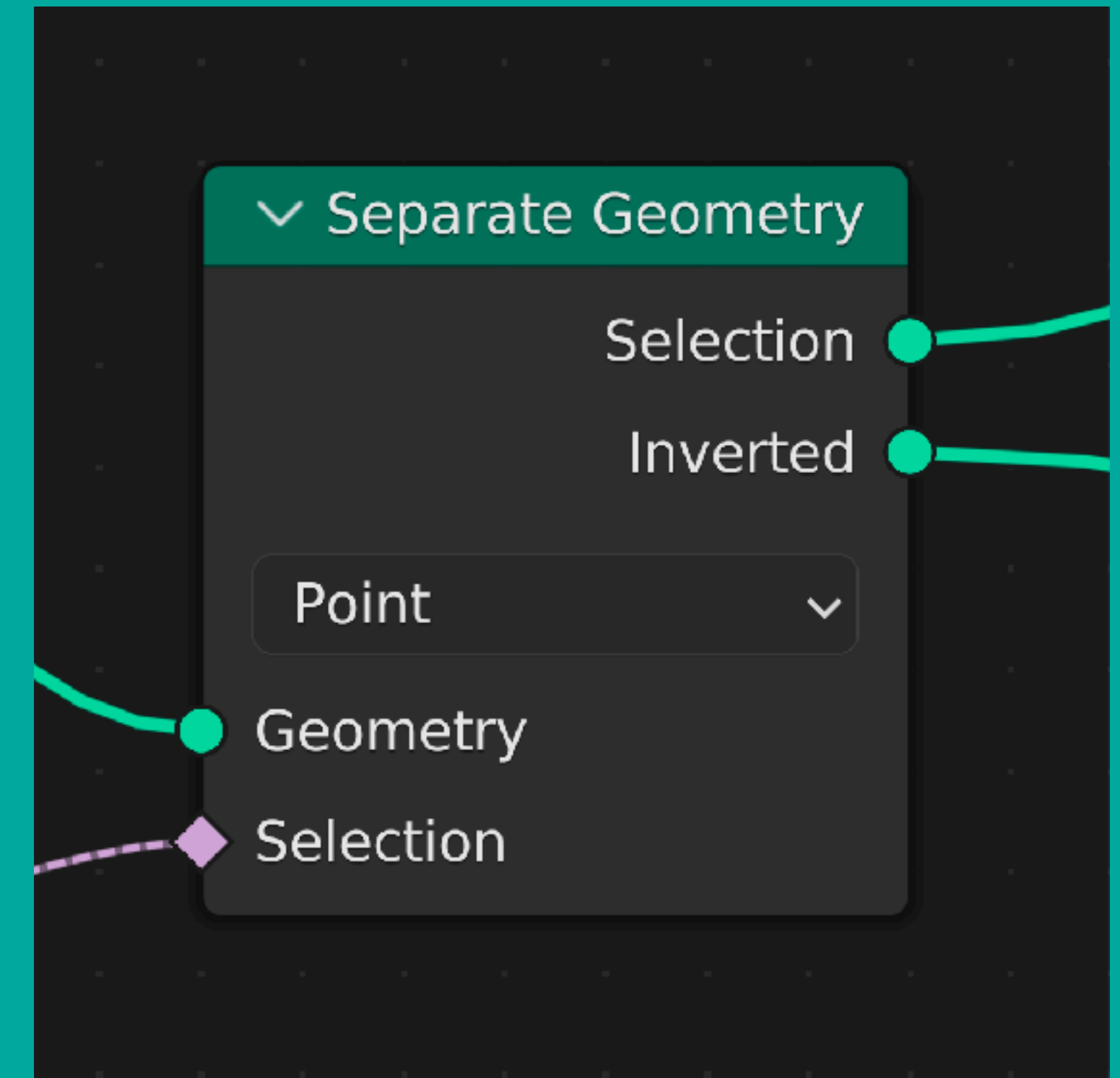
Distribute Points in Volume

- Fill a volume with points
- You can use Instance on Points after this
- Set to Grid if you want an even distribution, otherwise points will be randomly distributed



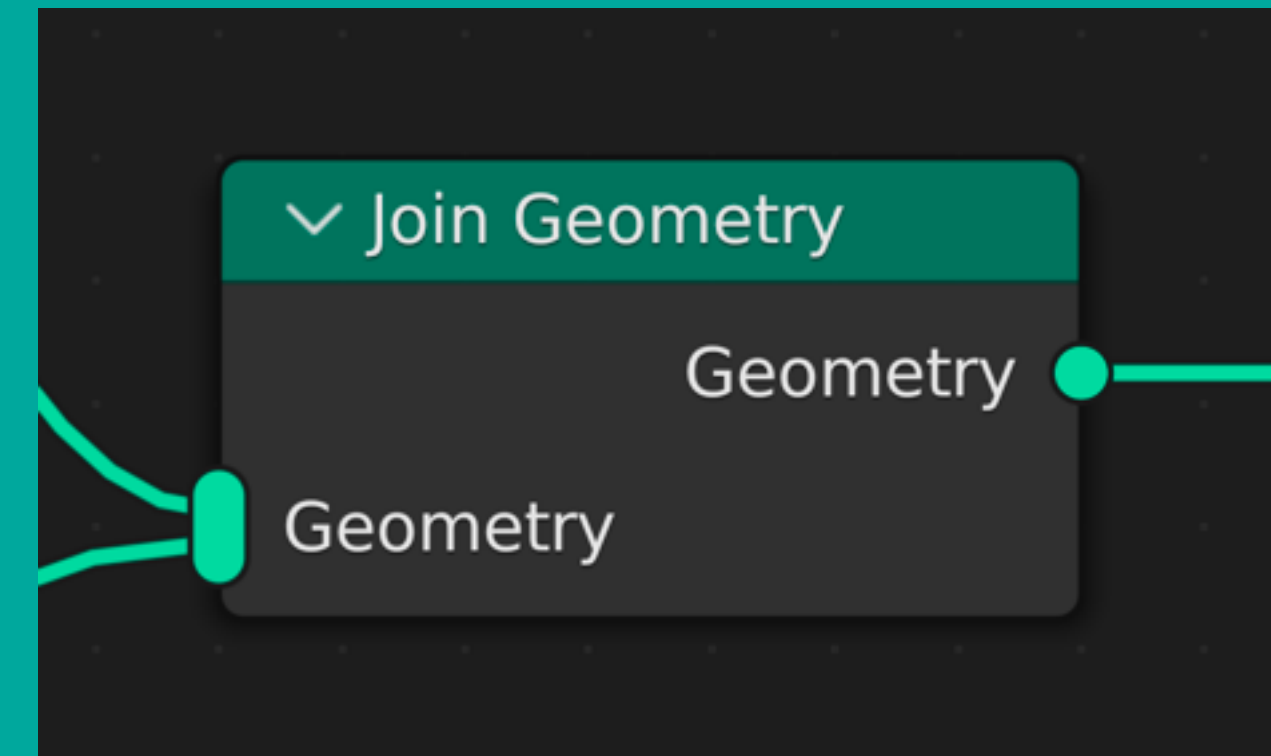
Separate Geometry

- Split the geometry in two
- Control the separation using the selection input



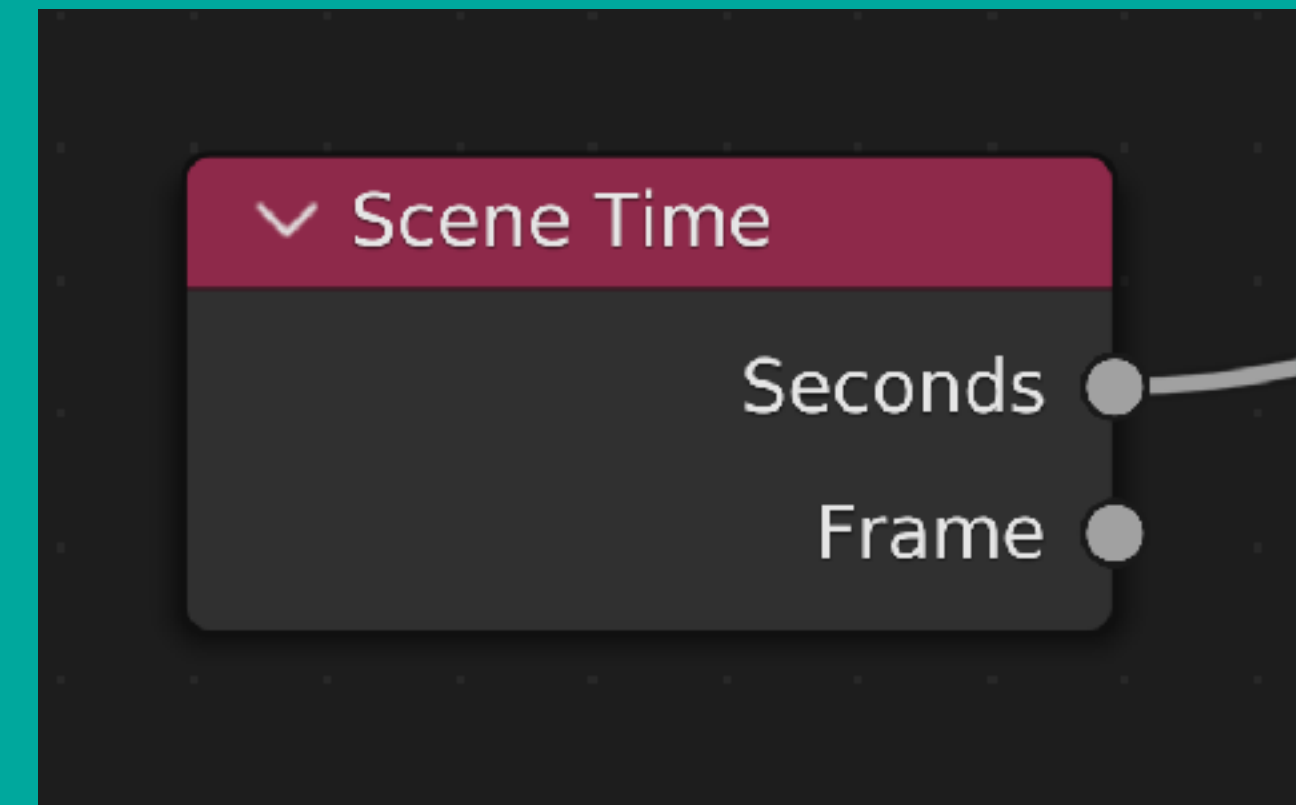
Join Geometry

- Always join Geometry back together before connecting it to Group Output



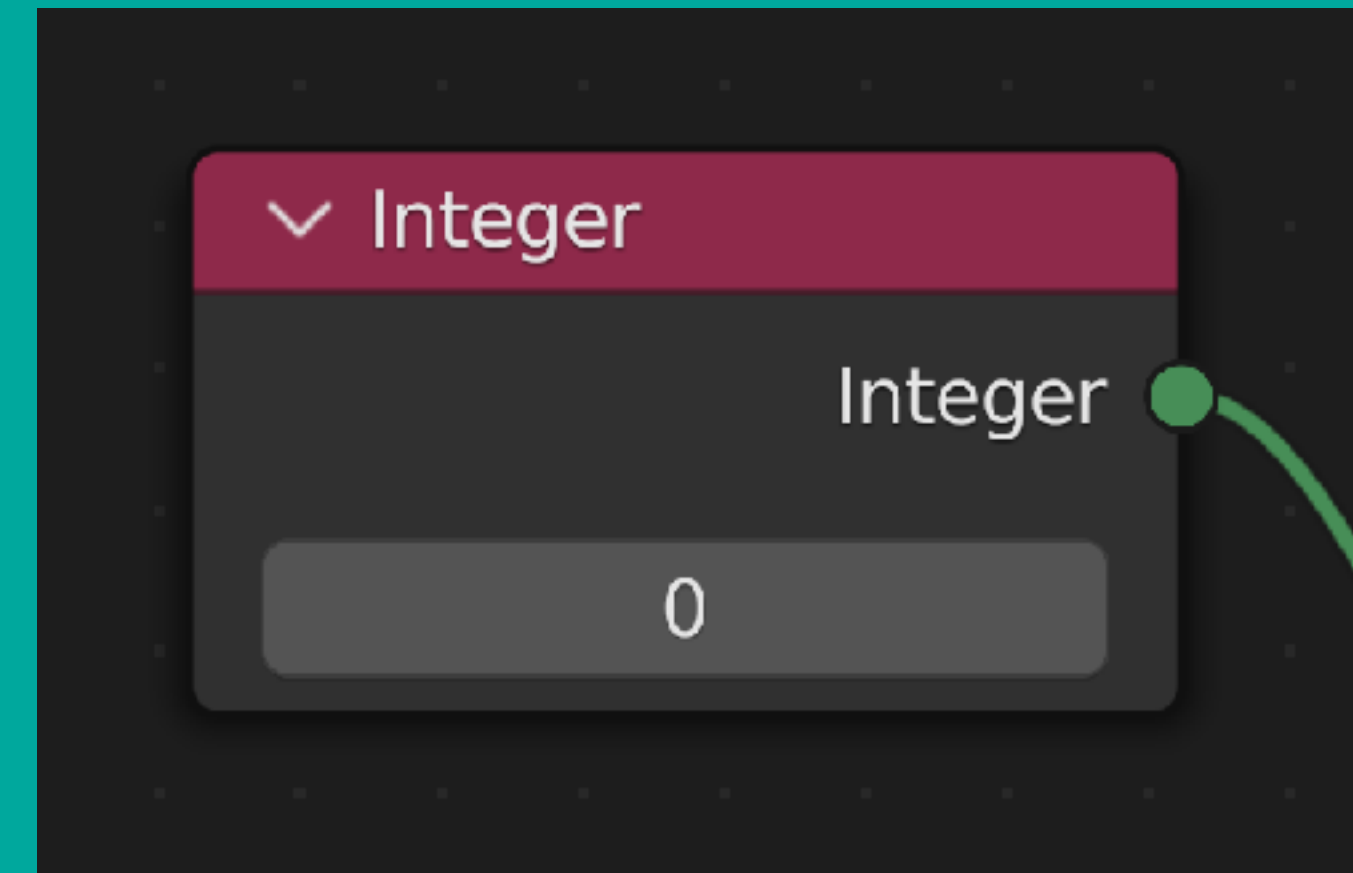
Scene Time

- Gives the position of the current animation
- So it outputs 0 when the playhead is at the beginning of the timeline
- If you use the Seconds output, it will output the current position of the playhead in seconds
- If you use the Frame output, it will output the current position of the playhead in frames
- The default frame rate in Blender is 24fps



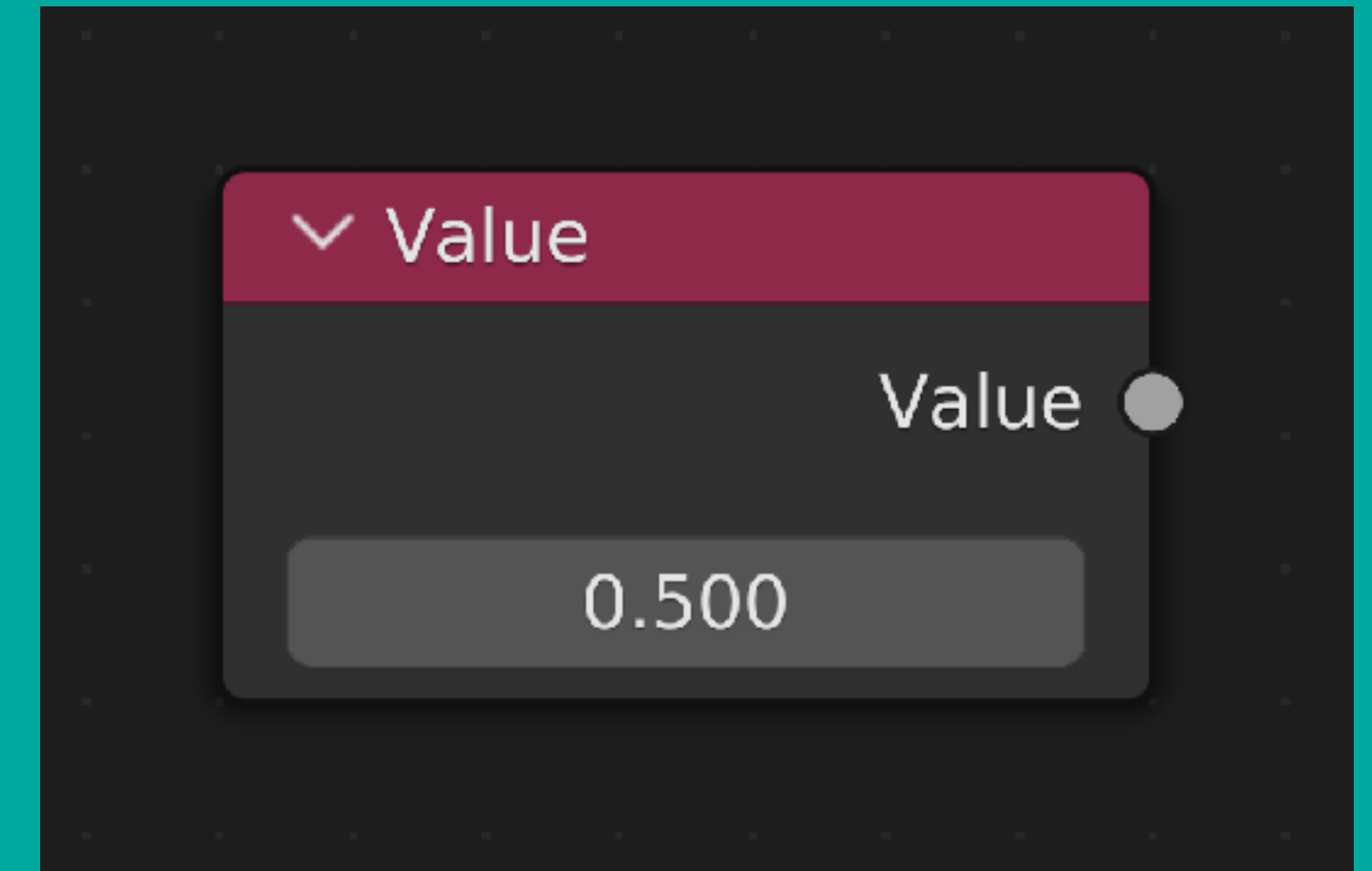
Integer

- Outputs a whole number that you can set
- Works like a constant



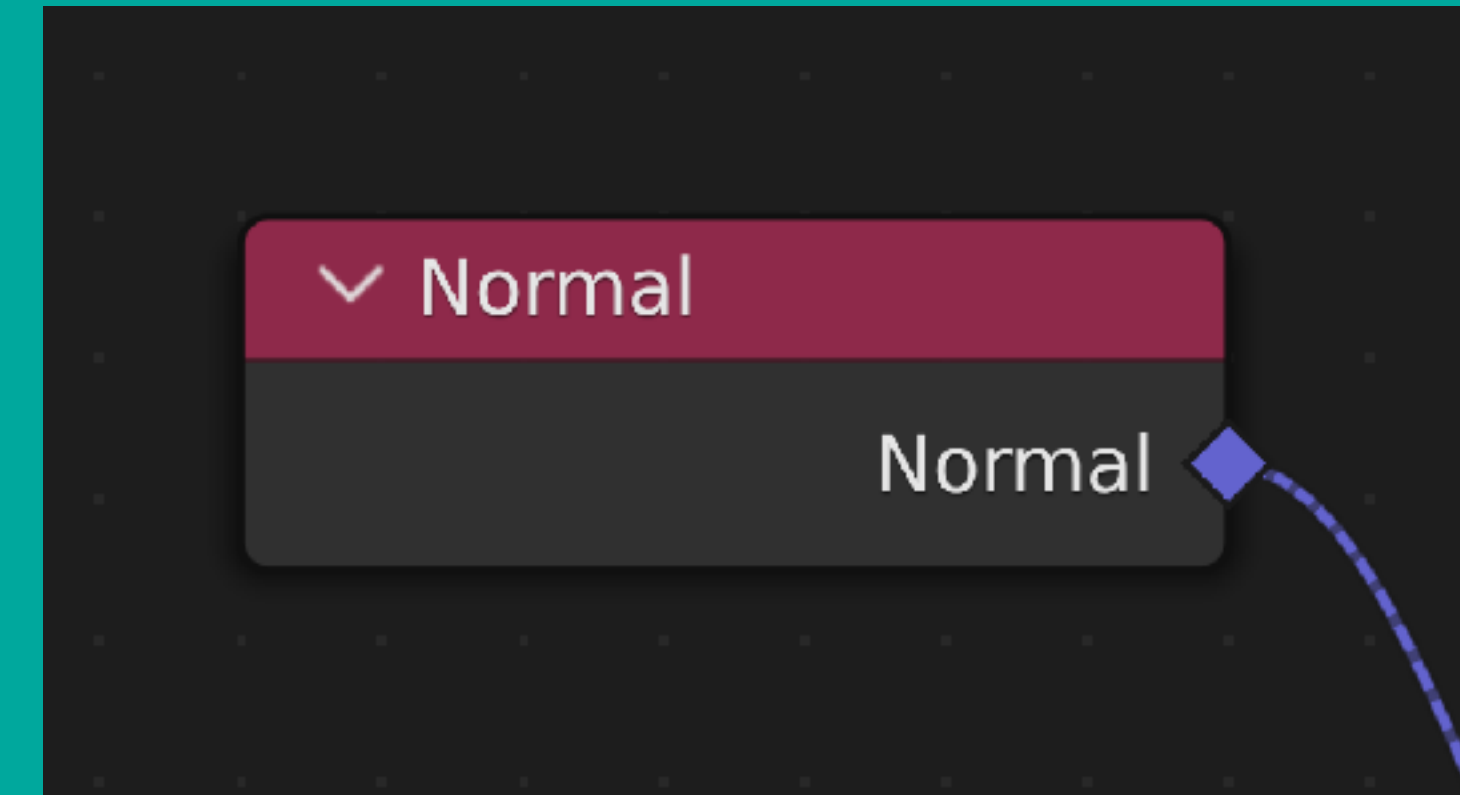
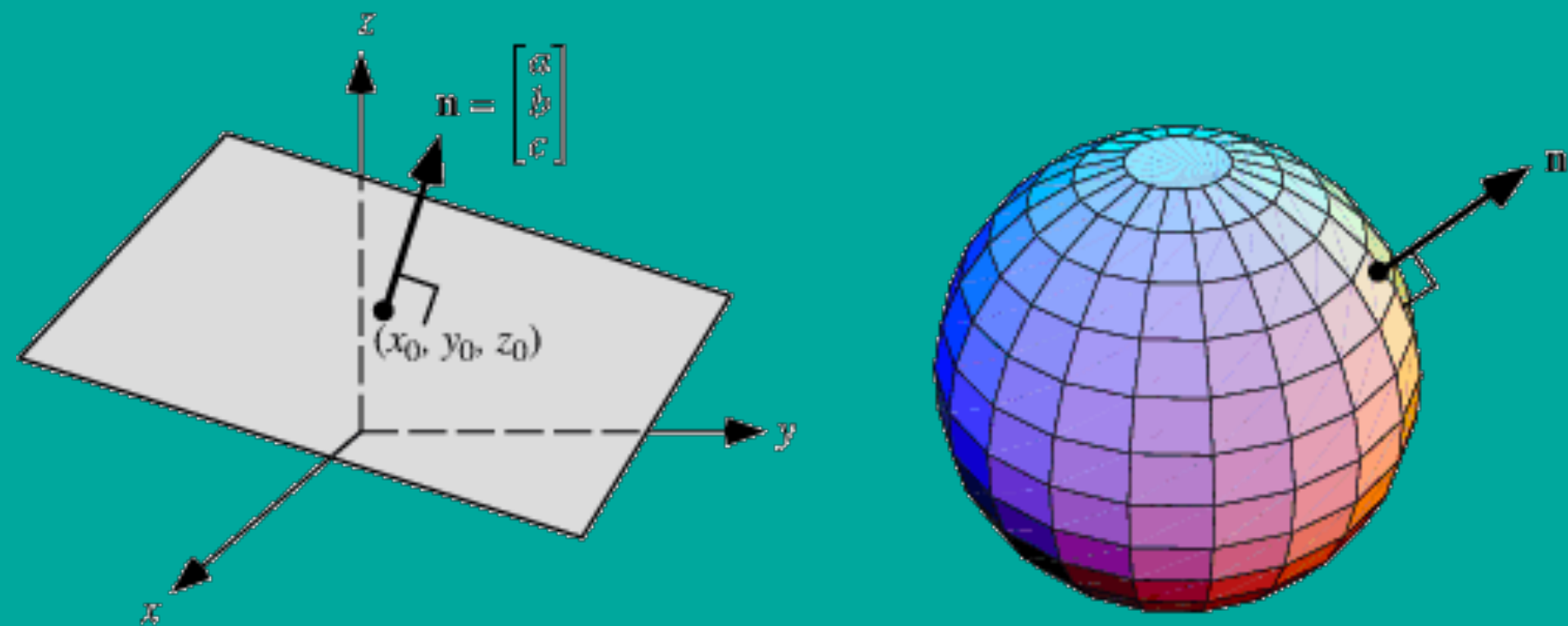
Value

- Outputs a decimal number (float) that you can set
- Works like a constant



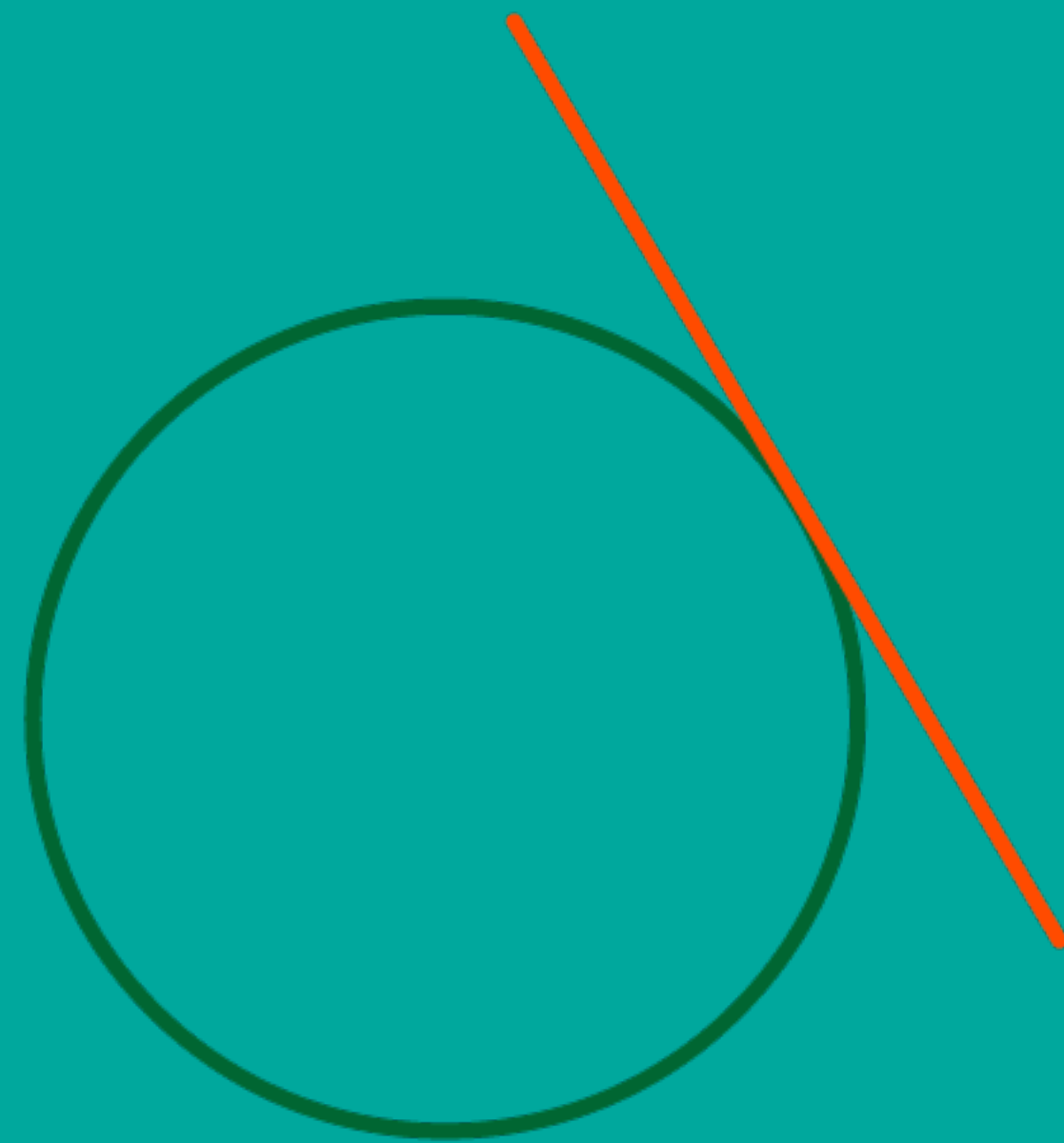
Normal

- Returns a vector which is perpendicular to the surface at a given point

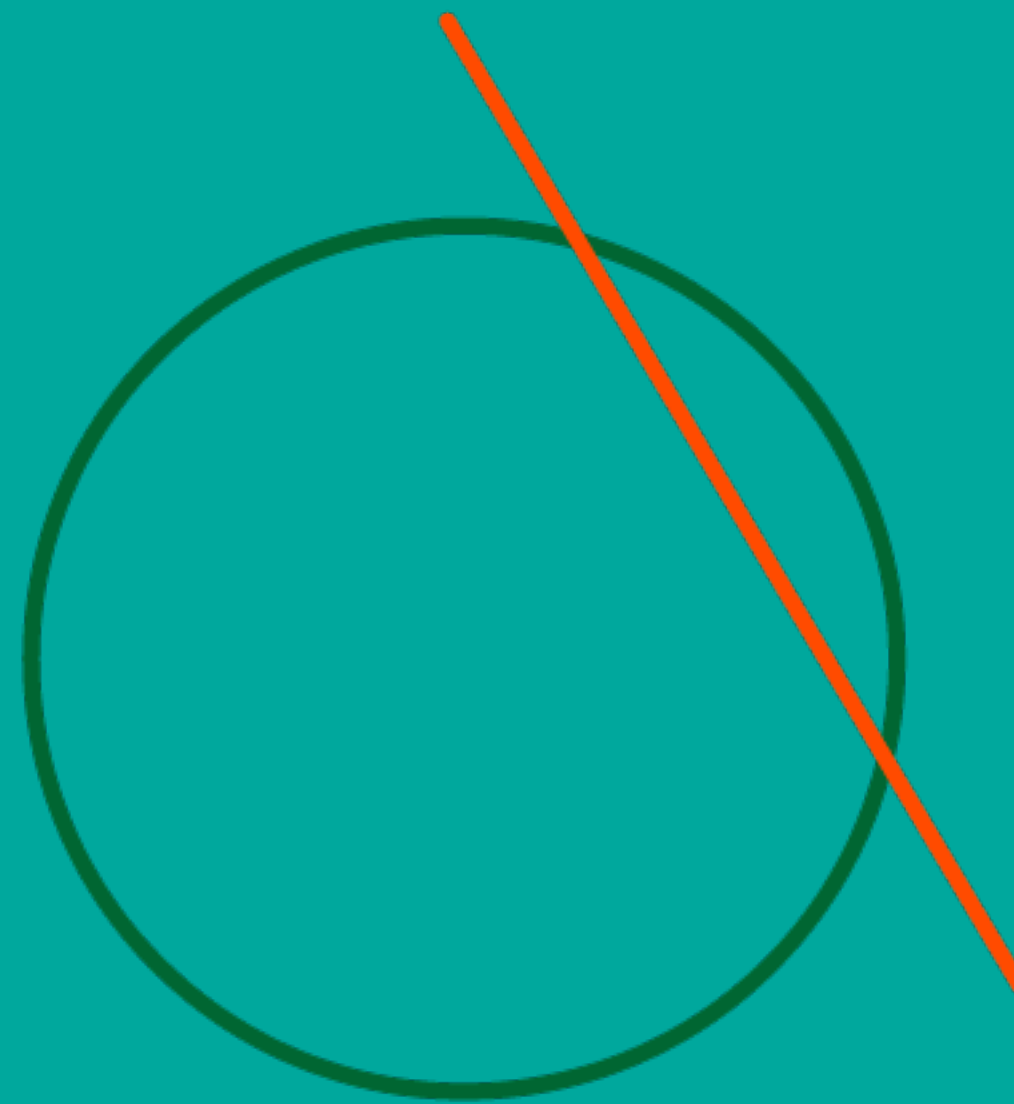


Curve Tangent

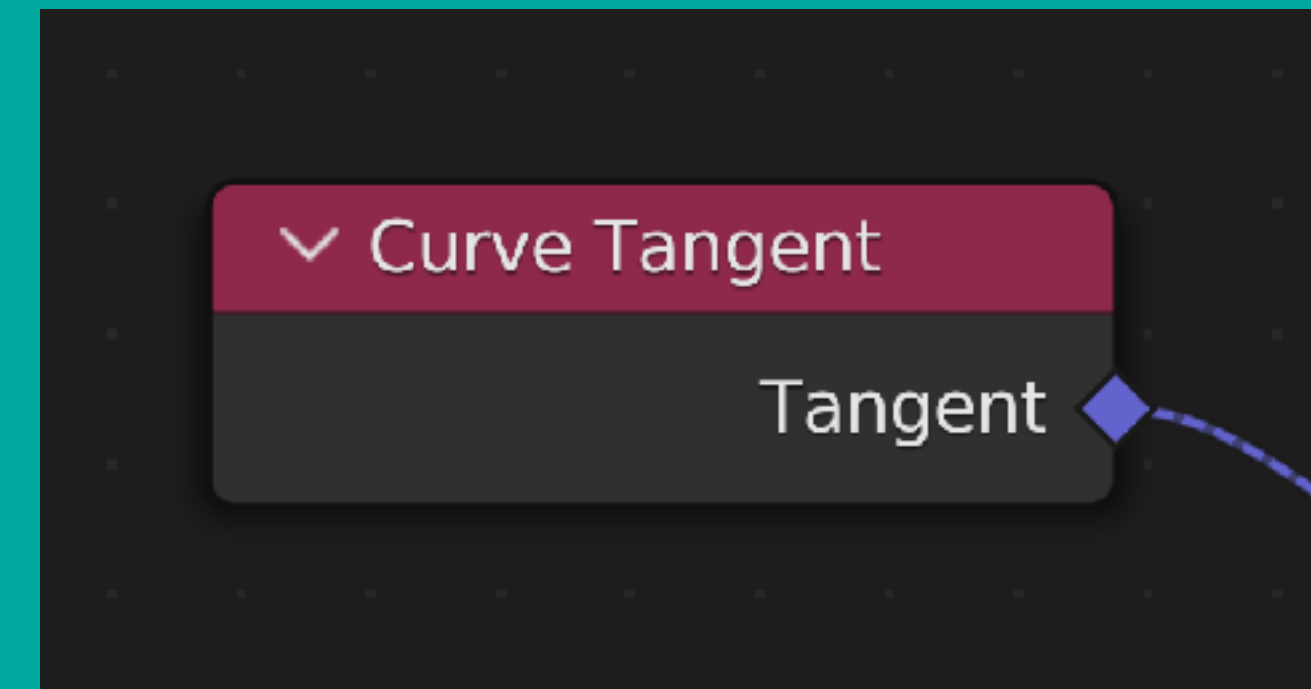
- Returns a vector which is the direction of the curve at a given point



Tangent

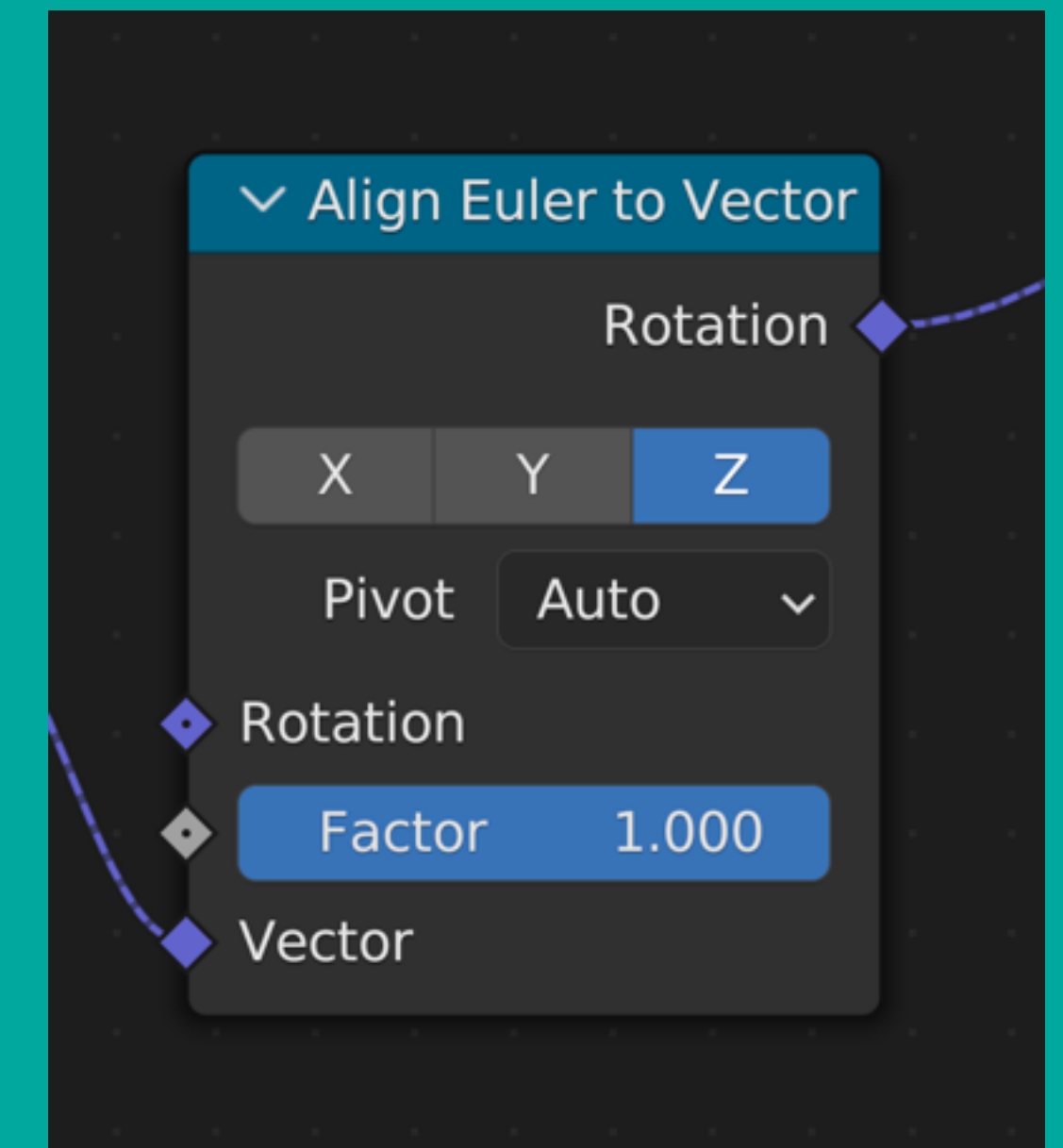
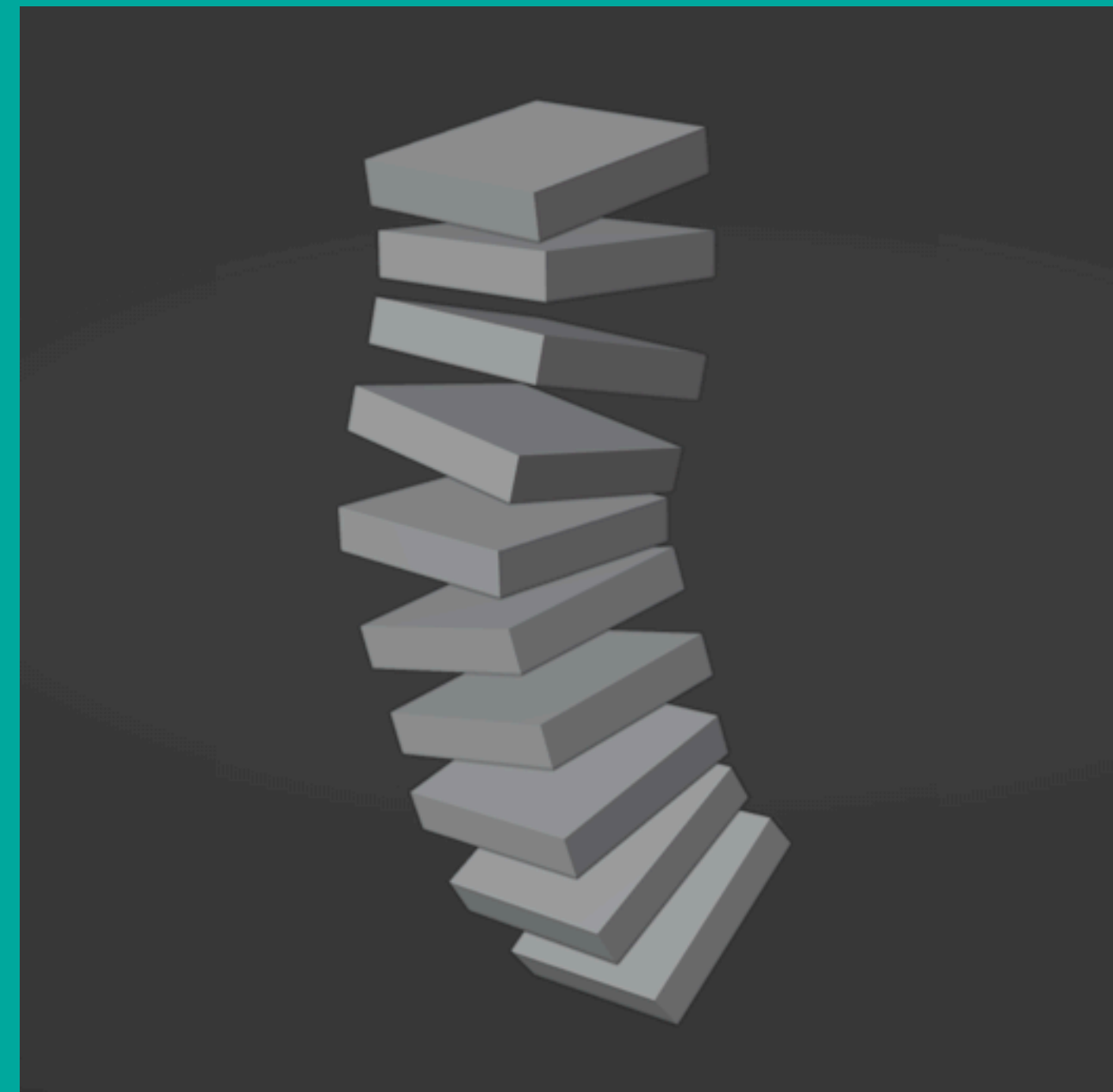
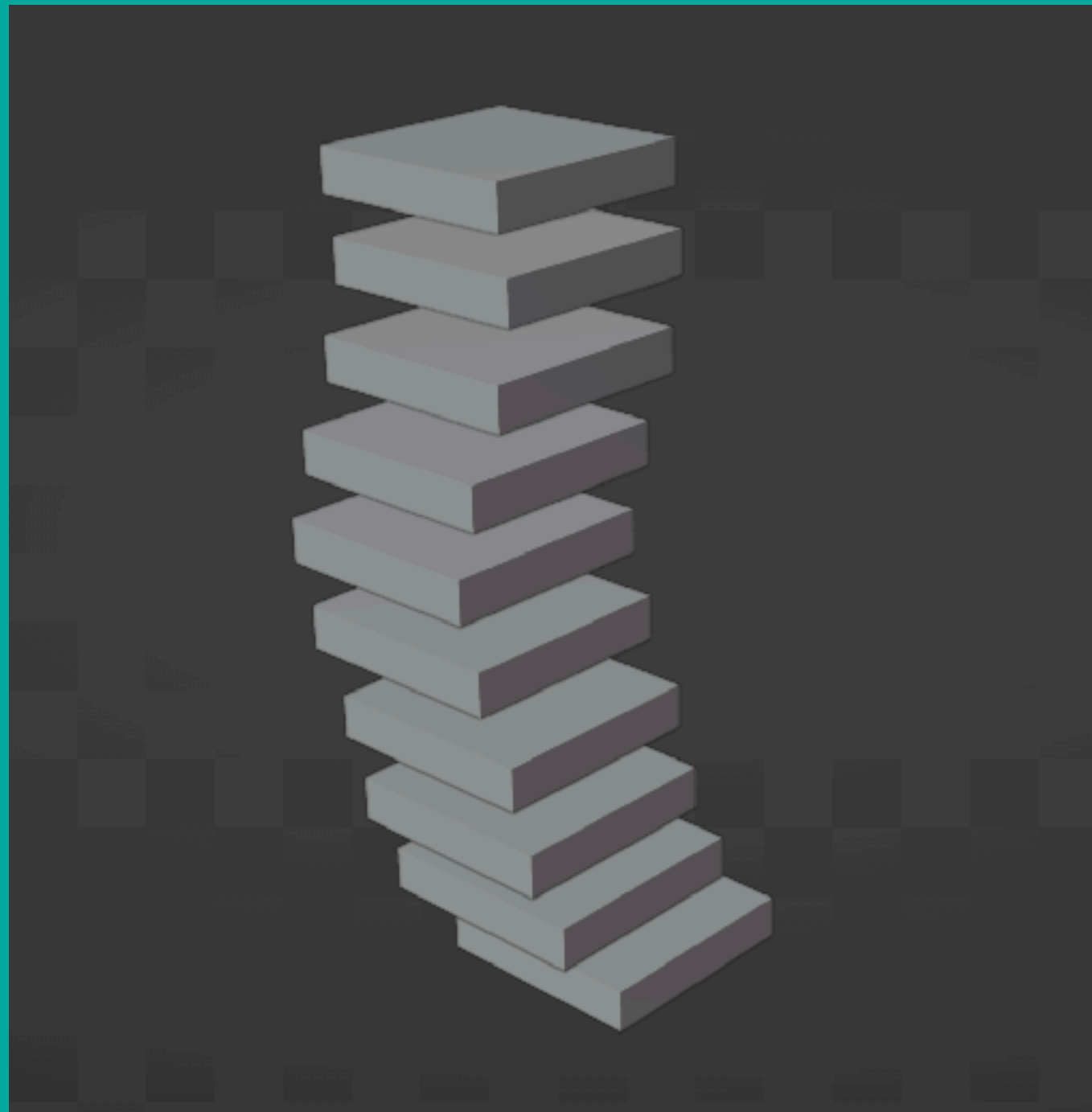


NOT Tangent



Align Euler to Vector

- Create a rotation that aligns the selected axis with the incoming vector



Node connections and their datatypes

